

GROWING UP WITH AN UNEMPLOYED MOTHER^{*}

Nataliya Gimpelson

Paulo Lins

International Monetary Fund Clemson University

This version:

May, 2026

ABSTRACT

We study the long-term consequences of maternal unemployment for children's labor-market outcomes. Using linked mother-child data from the National Longitudinal Survey of Youth, we document that children exposed to longer maternal unemployment spells have lower wages and employment probabilities as adults. We then unpack these relationships and investigate mechanisms. First, we show that these negative exposure effects are concentrated in adolescence. Second, we explore how mothers change their investment inputs. Third, we highlight how these relations persist after conditioning on family permanent income, suggesting mechanisms beyond income loss. Fourth, we use placebo tests and bunching diagnostics to address selection concerns. Overall, our results provide new evidence on the anatomy of the intergenerational impacts of unemployment and suggest important roles for non-income aspects of unemployment.

Keywords: Human Capital Investments, Unemployment

JEL - Classification: E24, J62

^{*}First draft: July, 2024. We thank Mark Bilz, Caitlin Hegarty, Jonathan Leganza, and David Slichter for their detailed comments on early drafts. We also appreciate the suggestions from Gregorio Caetano, Bill Dougan, David Drukker, Francisco Gallego, Aspen Gorry, Mike Makowsky, José Mustre del Río, Ronni Pavan, Daniel Ringo, Viviane Sanfelice, Scott Templeton, Paul Wilson, and participants of seminars at Clemson University, University of Rochester, and Midwest Macro-Purdue. The views in this paper are those of the authors and do not necessarily reflect the views of the IMF, its Executive Board, or its management. This research was conducted with restricted access to Bureau of Labor Statistics (BLS) data. The views expressed here are those of the authors and do not reflect the views of the BLS.

1 Introduction

Losing a job and becoming unemployed can be a traumatic and costly experience for workers and their families. A large literature shows that unemployment and displacement are associated with persistent income losses and declines in health among workers (Jacobson, LaLonde, and Sullivan, 1993; Stevens, 1997), and with worse outcomes for their children (Oreopoulos, Page, and Stevens, 2008; Britto, Melo, and Sampaio, 2023). Yet unemployment is a composite shock. It can reduce household resources, change parental time at home, and generate psychological and social costs. Moreover, the same unemployment spell may have different consequences depending on the child's developmental stage. A shock in early childhood, when parental time is a key input, may differ fundamentally from a shock in adolescence, when children are making educational investments and may be able to contribute financially to the household. Understanding the intergenerational cost of unemployment therefore requires asking not only whether parental unemployment matters, but when it matters and through which channels.

This paper documents and unpacks the association between maternal unemployment during childhood and children's long-run labor-market outcomes in the United States. We use linked mother-child data from the National Longitudinal Survey of Youth 1979 (NLSY79) and the NLSY79 Child and Young Adult survey (NLSY79-CYA). The NLSY79 records mothers' labor-force status on a week-by-week basis, allowing us to construct a cumulative measure of each child's exposure to maternal unemployment from ages 0 to 18. The NLSY79-CYA follows the biological children of the original cohort from childhood into adulthood, providing detailed measures of schooling, labor-force entry, wages, and employment. We relate maternal unemployment exposure to children's long-run outcomes, both cumulatively and by childhood stage, and then assess how the association changes as we condition on family permanent income and maternal background characteristics. In this sense, our exercise follows the spirit of Carneiro and Heckman (2003), who show that early-life conditions are strong predictors of later-life outcomes.

We begin by quantifying the overall relationship between maternal unemployment and labor-market outcomes. We find that children exposed to more maternal unemployment earn less as adults. A one-standard-deviation increase in exposure (about 67 additional weeks with an unemployed mother during childhood) is associated with roughly a 4% lower wage in adulthood, conditional on employment. This magnitude is economically meaningful: it corresponds to about 30-50% of the return to one additional year of schooling, which is typically estimated between 8 and 13% (Card, 1999). Exposure is also associated with a lower probability of employment in adulthood.

Next, we unpack this relationship through several additional exercises. First, we examine the role of education. Maternal unemployment exposure is associated with lower educational attain-

ment, consistent with a broader literature showing that early-life conditions shape human capital accumulation. [Carneiro and Heckman \(2003\)](#) document that family permanent income during childhood is a strong predictor of college enrollment; [Pedtke \(2026\)](#) and [Caucutt and Lochner \(2020\)](#) emphasize the role of family and income timing; and studies of tax credit expansions show that household resources during childhood predict educational outcomes ([Dahl and Lochner, 2012](#); [Agostinelli and Sorrenti, 2022](#); [Bastian and Lochner, 2022](#)). Our results suggest that maternal unemployment exposure is also in this set of early-life predictors of educational attainment.

Second, we examine whether these associations vary with the child’s age at exposure. We find that the negative relationship is concentrated in adolescence. Maternal unemployment exposure during early childhood (ages 0-5 and 6-12) has little predictive power for adult outcomes. By contrast, exposure during adolescence (ages 13-18) is a strong predictor of lower wages and employment in adulthood. We are the first U.S. study to document this age gradient; related evidence comes from [Carneiro, Salvanes, Willage, and Willén \(2024\)](#) for Norway, [Uguccioni \(2022\)](#) for Canada, and [Bingley, Cappellari, and Ovidi \(2024\)](#) for Denmark.

Third, guided by the age-gradient results, we use the American Time Use Survey (ATUS) and the Consumer Expenditure Survey (CE) to examine maternal time-use and expenditure patterns during unemployment. For early childhood, the results suggest that unemployed mothers increase time with children while reducing market inputs such as formal childcare and education spending. This pattern is consistent with input substitution largely offsetting the unemployment shock at young ages. Supporting this interpretation, early-childhood exposure does not predict lower test scores by age 10. For adolescence, the picture is different. Children exposed during this period enter full-time work earlier and attain less education. These patterns suggest that adolescent exposure predicts changes in human-capital accumulation and early-career decisions – channels for which additional maternal time is unlikely to compensate. Together, the results point to schooling and early-career choices as central channels of the intergenerational cost of unemployment, rather than early-childhood cognitive development.

Fourth, we assess how the associations between maternal unemployment and children’s outcomes change as we condition on family resources and other characteristics. Following [Carneiro and Heckman \(2003\)](#), we construct a measure of family per-capita permanent income during childhood and include it as a control variable. The associations shrink but persist. For example, conditional on permanent income, a one-standard-deviation increase in exposure is still associated with about a 3% lower adult wage. This result suggests that non-income aspects of unemployment have predictive power for adult outcomes beyond what income loss alone can account for. The estimates are also stable when we add rich controls for maternal background and ability proxies, family structure, and local labor-market conditions from the confidential NLSY Geocode data. Finally, throughout the paper, we distinguish unemployment from non-participation by constructing parallel exposure

measures, and show that the patterns are driven by unemployment exposure rather than by time out of the labor force, indicating that the results are not simply a “mother at home” effect.

Finally, we implement two diagnostic exercises to address selection concerns. First, we show that maternal unemployment outside the child’s ages 0-18 has little predictive power for adult outcomes. This provides a timing-based check against selection stories that operate uniformly over the mother’s life cycle. Second, we use a recently developed approach that exploits the bunching of observations at zero unemployment exposure to test for selection (Caetano, 2015; Caetano, Caetano, Fe, and Nielsen, 2025). We find that outcomes vary smoothly through zero, reducing concerns that the associations reflect selection on unobserved maternal traits.¹

Our results suggest that policies supporting unemployed workers may need to address not only contemporaneous income losses but also disruptions to children’s schooling and early labor-market transitions, particularly during adolescence.

Related Literature Our paper contributes to the literature on the intergenerational consequences of job loss and unemployment. Displaced workers suffer persistent earnings losses (Jacobson et al., 1993; Stevens, 1997; Couch and Placzek, 2010; Raposo, Portugal, and Carneiro, 2021), and a growing body of work traces these consequences to the next generation. The evidence spans multiple countries and outcomes: wage effects in Canada (Oreopoulos et al., 2008); labor-supply responses in Belgium (Fradkin, Panier, and Tojerow, 2019); school enrollment, teenage work, crime, and early pregnancy in Brazil (Britto et al., 2023); post-secondary enrollment in Canada (Coelli, 2011); infant health in the United States (Lindo, 2011); and school performance and grade retention in Norway and the United States (Rege, Telle, and Votruba, 2011; Stevens and Schaller, 2011). Other studies find more limited intergenerational consequences: small effects on college enrollment and early-career earnings in the United States (Hilger, 2016), and no significant effects on children’s earnings in Norway (Bratberg, Nilsen, and Vaage, 2008). We contribute to this literature by providing a unified account of how maternal unemployment during childhood predicts adult outcomes: when in childhood the association is strongest, how much of it is related to family resources, and how mothers adjust time and expenditure inputs during unemployment. The contribution has three parts.

First, we document how the predictive power of maternal unemployment exposure varies across childhood stages. We are the first to show this age gradient in the U.S., finding that the associations are concentrated in adolescence. This result is closest to Carneiro et al. (2024), who find that parental displacement in Norway matters more in early adolescence than in early childhood, with

¹Because exposure cannot be negative, the zero-exposure bin mechanically over-represents mothers with strong labor-market attachment. If the baseline associations were driven by unobserved maternal traits that separate never-unemployed mothers from otherwise similar mothers with minimal unemployment exposure, outcomes should jump discontinuously at zero.

effects operating partly through children’s and parents’ mental health rather than income alone. It contrasts with [Uguccioni \(2022\)](#), who finds larger effects of parental unemployment in Canada for children exposed before age 10, and with [Bingley et al. \(2024\)](#), who find that parental job loss in Denmark is most consequential when children are exposed in infancy.

Second, we distinguish the income and non-income channels underlying the association. We focus on cumulative unemployment exposure rather than discrete displacement events. Prior work has explored mechanisms through subgroup heterogeneity and non-labor outcomes such as child health ([Lindo, 2011](#); [Carneiro et al., 2024](#)), school performance ([Coelli, 2011](#); [Rege et al., 2011](#); [Stevens and Schaller, 2011](#); [Britto et al., 2023](#)), and early labor-market behavior ([Fradkin et al., 2019](#)). Because displacement simultaneously lowers income and imposes other costs, separating these channels is difficult. Our cumulative exposure measure, combined with a permanent income control following [Carneiro and Heckman \(2003\)](#), provides a way to assess the relative importance of income and non-income channels.²

Third, we document how maternal unemployment changes household inputs. This contribution connects our age-gradient results to research on parental time, household resources, and child development. Our focus on unemployment exposure complements [Krueger and Mueller \(2012\)](#), who study time allocation among unemployed workers more broadly, and [Blau and Grossberg \(1992\)](#), who examines maternal time investment using the share of weeks mothers worked during early childhood. It also relates to work on the income-time trade-off in child development, including studies of maternity leave expansions ([Dustmann and Schönberg, 2012](#); [Carneiro, Løken, and Salvanes, 2015](#)), maternal employment and childcare ([Bernal, 2008](#); [Baker, Gruber, and Milligan, 2019](#)), and the effects of maternal labor supply on children’s cognitive development ([Caetano, Caetano, Nielsen, and Sanfelice, 2024](#)).

Roadmap Section 2 describes our empirical approach. Section 3 introduces our data sources and measurement strategy. Section 4 presents our main findings. Section 5 examines the underlying mechanisms. Section 6 concludes.

²Similar to [Hilger \(2016\)](#), we find evidence that part of the association reflects selection on unobservables. We therefore interpret the estimates as predictive associations rather than causal effects. Still, placebo regressions and bunching tests ([Caetano et al., 2025](#)) provide diagnostics for some selection concerns. The placebo exercises address selection that predicts outcomes uniformly over the mother’s life cycle. The bunching test exploits the fact that zero unemployment exposure mechanically concentrates mothers with especially strong labor-market attachment. If unobserved traits associated with this high attachment drive the baseline association, children’s outcomes should jump at zero relative to very small positive exposure. If exposure itself matters, outcomes should vary smoothly through the boundary.

2 Conceptual Framework and Empirical Strategy

In this section, we motivate and describe our empirical strategy for documenting the association between mothers' unemployment and their children's long-term outcomes. For ease of interpretation, we use linear models for our main analysis. Appendix B.2 examines whether the associations are in fact non-linear.

2.1 Conceptual Framework

There are three main channels through which maternal unemployment may shape children's long-run labor-market outcomes. First, unemployment reduces household income, thereby limiting the market inputs available for investments in children's human capital. Second, it increases mothers' time at home, which could partly offset the loss of financial resources if that time is allocated to activities that build children's human capital. Third, maternal unemployment may operate through other non-income channels, such as stress and depression experienced by mothers or children, other health-related concerns, family instability, or social stigma.

The relative importance of these channels is likely to vary with the child's age at exposure. In early childhood, parental time is a central input into child development, and mothers may be able to substitute additional time for reduced market inputs such as formal childcare and educational spending. If this substitution is effective, the association between maternal unemployment and children's outcomes may be weak at young ages. By contrast, during adolescence, when schooling and early labor-market decisions are being made, the relevant investments may be less easily replaced by parental time. Adolescents may also respond directly to household financial pressure by entering the labor force earlier or reducing educational investments. These considerations suggest that the predictive power of maternal unemployment for adult outcomes may be concentrated in adolescence rather than early childhood. Whether this pattern holds is an empirical question.

Most existing work studies parental job-displacement events, often exploiting variation from plant closures. This approach is well suited to estimating the overall consequences of job loss, but the event combines several changes that occur simultaneously: lower income, changes in parental time, and other psychological or social costs. We take a complementary approach by measuring children's cumulative exposure to maternal unemployment from ages 0 to 18. This measure captures the intensity of unemployment exposure over childhood and can be decomposed by childhood subperiod, allowing us to study when exposure is most predictive of adult outcomes. To speak to mechanisms, we then examine how the association changes after conditioning on family permanent income. This comparison provides evidence on the extent to which the relationship is accounted for by long-run family resources rather than other aspects of unemployment.

2.2 Baseline Specification

We estimate the association between exposure to maternal unemployment and labor market outcomes by projecting young adult i 's outcome at time t , y_{it} , on measures of maternal unemployment and out-of-the-labor-force exposure:

$$y_{it} = \alpha + \beta_1 UNEMP_i + \beta_2 OLF_i + \gamma_1 X_i + \gamma_2 Z_{it} + \epsilon_{it} . \quad (1)$$

y_{it} denotes a young adult's labor market outcome. $UNEMP_i$ and OLF_i capture cumulative childhood exposure to maternal unemployment and out-of-the-labor-force spells, respectively. X_i and Z_{it} include time-invariant and time-variant child characteristics, and ϵ_{it} is an error term. We include OLF exposure to ensure that the coefficient on unemployment is not confounded by broader maternal non-employment; our results are stable whether or not we condition on OLF. In Section 3, we provide detailed variable definitions and, in Section 5, we augment this specification by conditioning on family permanent income to assess the extent to which the associations persist after accounting for family resources.

The timing of the variables is central. Exposure measures are calculated before age 18, while outcomes are measured when the child reaches working age. For those with a high school degree or less, this would be older than 21 years, and for those with a college degree, it would be older than 25 years. Thus, equation (1) is best viewed as a forecasting exercise in which early-life conditions predict later-life outcomes, in the spirit of [Carneiro and Heckman \(2003\)](#), who showed that family permanent income is a strong predictor of children's future educational attainment.

We estimate equation (1) by OLS and interpret β_1 as the partial correlation between childhood exposure and adult outcomes conditional on observables – how much a unit increase in maternal unemployment exposure predicts worse adult outcomes, regardless of whether that increase falls within a longer spell or an isolated short spell.³ Throughout the paper, we interpret these estimates as reduced-form associations conditional on observables. Specifications that additionally control for permanent income or educational attainment should be understood as changing the conditioning set and therefore targeting different conditional associations rather than the same underlying estimand.

Because cumulative unemployment exposure is not randomly assigned, the main empirical concern is that it may correlate with unobserved determinants of children's outcomes. This concern is sharper for OLF, which often reflects active choices correlated with unobserved maternal characteristics, than for unemployment spells driven by involuntary job loss.⁴ For this reason, our

³Appendix B.5 decomposes exposure into the number of unemployment spells and the average duration per spell. The results suggest that the frequency of spells is a more important driver of the negative association with children's future wages.

⁴To make the coefficients of OLF more comparable with those of UNEMP, we report in Appendix B.10 an IV strategy

analysis focuses on unemployment exposure, and we complement it with diagnostic tests to assess how far selection on unobservables can account for the baseline patterns.

3 Data and Measurement

In this section, we describe the dataset and measurement choices used to study the association between maternal unemployment and children's long-term outcomes. The data provide detailed information on mothers' labor market status and follow their children into adulthood, allowing us to link maternal unemployment histories to their offspring's labor market outcomes. Our key measure is cumulative exposure to maternal unemployment during childhood.

3.1 The NLSY79 and NLSY79-CYA

We use data from two National Longitudinal Surveys.

First, we use the National Longitudinal Survey of Youth 1979 (NLSY79). The NLSY79 consists of a nationally representative sample of over 12,000 individuals ages 14-22 in 1979. The survey follows individuals longitudinally and has information about their employment, education, family background, and other life circumstances. Importantly for our exercise, the NLSY79 keeps a detailed record of each respondent's employment, unemployment, and non-employment spells at a weekly frequency. It has records from January 1, 1978, to the date of the last interview. We use this labor force status array to construct our measure of a child's exposure to maternal unemployment, which will be described in the following subsection.

Second, we use data from the NLSY79 Child and Young Adult (NLSY79-CYA). The NLSY79-CYA has information on children born to women in the original NLSY79 cohort. These children were born between 1980 and 2016 and followed longitudinally from birth through adulthood. It also includes extensive information on these children, including cognitive, socioemotional, and physical development. It also includes information on their educational attainment, employment status, and earnings by following children into their adulthood.⁵

In our analysis, we focus on children of mothers from the original NLSY79 cohort who have reached working age. The final sample consists of respondents who were interviewed at least once

using industry cyclicality to capture involuntary non-employment. The resulting point estimates are consistent with our main results.

⁵We construct hourly wages for NLSY79-CYA respondents following the methodology used by the Bureau of Labor Statistics for the main NLSY79 cohort. While BLS provides constructed hourly wage variables for NLSY79 and NLSY97 respondents, this variable is not available for children in the NLSY79-CYA. We replicate the BLS procedure documented in the HRP:PROC code described in the NLSY79 Codebook Supplement, Addendum to Appendix 18: Work History Data (page 318).

as adults. We further restrict the sample to young adults who are at least 21 years old with a high school diploma or less, or at least 25 years old with a college degree or higher. These restrictions yield a sample of 6,861 individuals born between 1980 and 1997, observed in the labor market between 2005 and 2016, and with an age range of 21–35. On average, respondents are observed 3.3 times, yielding a total of 22,920 respondent–year observations. Among the employed, 4,276 respondents provide non-missing wage information, with an average of 2.03 observations per person, resulting in 8,682 wage observations. Each respondent is linked to their mother. Appendix A reports summary statistics and further details on sample construction.⁶

We study a range of adult outcomes: log earnings, log wages, log hours worked, and indicators for educational attainment and labor force participation. The NLSY records all jobs ever held, so respondents may report multiple jobs in a given year. In our baseline specification, we select the main match, defined as the job with the most hours at the interview date, and pool observations across all survey waves, creating a panel where individuals contribute multiple person-year observations. Appendix B1 shows that our results are robust to alternative sample selection criteria.⁷

Our main specification relates variables measured during childhood to outcomes in adulthood. Control variables observed repeatedly over the 18 years of childhood are aggregated using the mode for categorical measures (e.g., marital status) and the mean for continuous measures (e.g., spouse’s workweek).

3.2 Constructing a Measure of Exposure to Maternal Unemployment

We measure children’s exposure to maternal unemployment as the fraction of their childhood that their mothers spent unemployed. This measure captures the intensity of mothers’ unemployment, distinguishing between those who lost a job once and immediately found a new one and those who have been consistently unemployed. For example, a single, short spell means a small fraction of childhood spent with an unemployed mother, while multiple, long spells mean a larger fraction. Previously, the literature has mainly looked at the effects of parental job loss in event study settings.

⁶We use the NLSY instead of the Panel Study of Income Dynamics (PSID) for two main reasons. First, the NLSY provides a detailed weekly array of labor market statuses, the key feature we rely on to create our unemployment exposure measure. The PSID measures unemployment by asking respondents how many weeks they were unemployed the previous year. Second, the NLSY offers detailed data on children’s backgrounds during their childhood and their labor market outcomes in adulthood. This second reason may be less critical since the PSID includes two supplements, the Child Development Supplement (CDS) and the Transition into Adulthood Supplement (TAS), which could, in principle, allow similar analysis.

⁷Our sample construction raises a potential concern: because maternal unemployment exposure predicts lower educational attainment, exposed children may enter the adult sample earlier and therefore be observed at different ages and levels of labor-market experience than less-exposed children. This could mechanically affect observed wages and employment outcomes. To assess robustness, we implement four alternative sample selection methods: (1) using only the first observed main match, (2) using only the last observed main match, (3) including all jobs across all surveys, and (4) averaging job characteristics within each survey year. Appendix B1 shows that the results remain similar across these alternative constructions.

Our approach complements the previous literature by using a different measure that captures the intensity of unemployment.

To compute the fraction of children’s childhood that their mothers spent unemployed, we explore the detailed labor force status array recorded in the NLSY79. It contains weekly information on the respondent’s labor market status, whether “employed,” “unemployed,” “out of the labor force,” or “on active military service.” In the case of employment, it also recodes some job characteristics. In each interview, respondents are asked to report their labor market status for every week since their last interview, which means that the weekly data covers the entire time the respondent participated in the survey, even including years when they were not interviewed, and provides a comprehensive work history for each respondent.

More formally, we define the exposure of child i to his/her mother’s unemployment as:

$$Exposure\ to\ Unemployed_i = \frac{\sum_{\{t|age_{i,t} \in [0,18)\}} \mathbb{1}\{labor\ status_{mother(i),t} = unemployed\}}{\sum_{\{t|age_{i,t} \in [0,18)\}} \mathbb{1}\{week_{mother(i),t} = observed\}}. \quad (2)$$

In the numerator, we count the number of weeks during which the mother reported being unemployed in the labor force status array. Here, $mother(i)$ is a function that denotes the identifier of child i ’s mother, and $labor\ status_{mother(i),t}$ is a variable indicating her labor market status in week t . The summation is carried out over $\{t|age_{i,t} \in [0,18)\}$, which represents all weeks in the first 18 years of life, from birth up to the eighteenth birthday. In the denominator, we count the number of weeks during which the mother’s labor force status is observed. $week_{mother(i),t}$ is a variable that indicates whether the labor market status of the mother in week t is observed.⁸

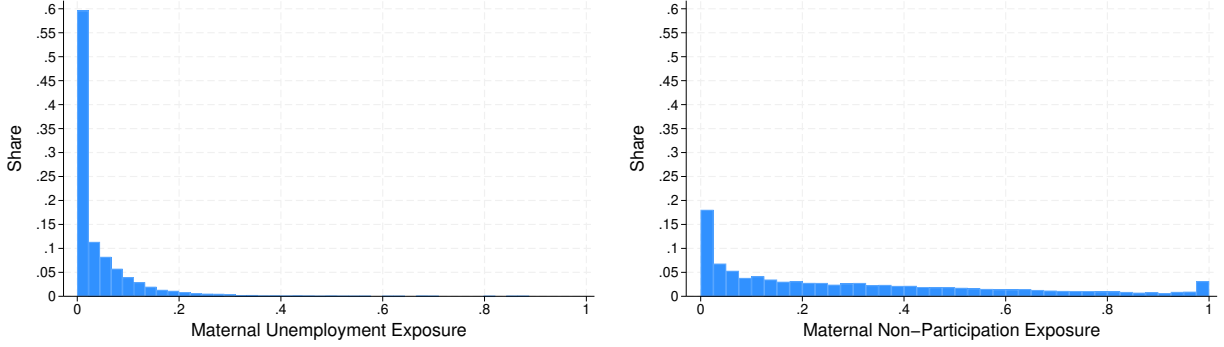
Following the same logic, we construct other exposure measures capturing the fraction of a child’s childhood in which his/her mother was out of the labor force or employed. The three constructed measures of exposure sum to one by definition. [Blau and Grossberg \(1992\)](#) also use the share of weeks that mothers worked as a measure of the quantity of maternal time available to invest in children. However, we focus on unemployment, which is likely involuntary, unlike out-of-the-labor-force status that reflects choice.

Figure 1 shows the distributions of exposures to mothers being unemployed and out of the labor force. There is much more variation in the out-of-the-labor-force exposure than in unemployment.

⁸We implement our measure by first calculating maternal unemployment exposure for each year of a child’s childhood. For instance, we count unemployed weeks and total observed weeks when the child was between ages 0 and 1, 1 and 2, and so on through ages 17 and 18. Then, we compute the ratio between unemployed weeks and total observed weeks for each childhood year. We have a total of 206,436 child-year observations. We treat years with 50 or fewer observed weeks as missing, which accounts for only 0.46% of child-year observations (948 in total), all of which occur when children are ages 0–18 before 1978 or after 2018, reflecting the fact that labor force status data are only available starting in 1978 and our sample extends through the 2019 release. As the last step, we average these ratios over the 18 years of childhood to construct a measure representing the exposure to maternal unemployment.

During the first 18 years of their lives, young adults with at least one wage observation were, on average, exposed to 1 year of maternal unemployment, to 6.8 years of their mother being out of the labor force, and to 9.9 years of maternal employment. Standard deviations are 1.4 years for the exposure to unemployment, 5.3 years for the exposure to a mother being out of the labor force, and 5.6 years for the exposure to maternal employment.

Figure 1: Fraction of Time in Each Labor Market State



Note: Maternal unemployment and non-participation exposures are defined as in equation (2) and can take values from 0 to 1. Each bar represents the share of young adults who were exposed to a particular level of maternal unemployment/non-participation during the first 18 years of their lives.

3.3 Measuring Family Permanent Income

We construct a measure of family income during childhood following [Carneiro and Heckman \(2003\)](#), from whom we borrow the name *family permanent income*. In Section 5, we use it to assess the extent to which the association between maternal unemployment and children’s outcomes persists after conditioning on family resources.

More formally, we measure family income by constructing a discounted average of family per capita income over the child’s childhood. The child i ’s family permanent income is defined as:

$$PI_i = \sum_{\{t | \text{age}_{i,t} \in [0,18)\}} \frac{Y_{i,t}}{(1+r)^t} \cdot \frac{\frac{1}{1+r} - 1}{(\frac{1}{1+r}^{19} - 1)} . \quad (3)$$

$Y_{i,t}$ is a measure of the per-capita income of i ’s family at time t . Unlike the exposure measure in equation (2), which is constructed from weekly labor-force histories, permanent income is measured at the annual frequency. We use a constant interest rate, r , to express income in terms of the year the child was born.⁹ Lastly, we sum income for all the child’s childhood years and divide it by the sum of the discount factors to compute average income. (In other words, we normalize the weights to sum to one.)

⁹We calibrate $r = 0.05$, but our main results are robust to other values.

Our measure of per capita income is net family income divided by the family size. The BLS created the total net family income variable by summing different income sources for all household members. These include, for example, labor income as wages and salary, farm and business income, asset income, and government transfers. We manually create a family income measure for respondents for whom this variable is unavailable. The BLS also created the variable family size by counting the number of people who live in the household and are related to the respondent by blood, marriage, or adoption.¹⁰ All values are expressed in \$10,000 in 1993 US Dollars, using the CPI. In our sample of young adults with non-missing wage observations, the average family permanent income is 0.865, with a standard deviation of 0.897.

Ideally, we would have the BLS's family income for each year of a child's childhood. However, this is not the case for two reasons. First, the NLSY began as an annual survey but transitioned to a bi-annual schedule after 1994. Therefore, by survey design, we do not observe family income for all the years of a child's childhood. Second, if a respondent misses a survey, he/she is not asked about income in the missed years. On average, we observe income for 10.17 years with a standard deviation of 4.03 years. We explain below how we correct for measurement error in our results.

3.4 Selection on Time-Invariant Unobservables

A concern with our exposure measure is that it may capture latent maternal characteristics rather than the predictive content of maternal unemployment. Mothers with weaker non-cognitive skills, lower emotional stability, or less effective parenting styles may both invest less in their children's human capital and experience higher unemployment. We address this concern in three ways. First, we control for proxies of maternal quality, including educational attainment and Armed Forces Qualification Test (AFQT) scores. More educated mothers and those with higher AFQT scores tend to invest more effectively in their children and face lower unemployment risk, so including these proxies absorbs part of the variation in unobserved maternal traits. We then show in a separate robustness exercise that the results are also stable to including a broad set of prenatal and birth characteristics that predict later outcomes (Dogan, García, and Polovnikov, 2025).

Second, we implement a placebo test using maternal unemployment outside the child's formative years. This exercise addresses selection stories in which unemployment exposure primarily proxies for persistent maternal traits that predict children's outcomes regardless of when unemployment occurs. If such timing-invariant selection were driving the baseline results, unemploy-

¹⁰When calculating family size, the BLS only counts family members related to the respondent by blood, marriage, or adoption. Foster relationships, partners, boarders, guardians, and others are not considered. Similarly, the income and earnings of spouses are included in the total family income, but those of partners are not. The reason for this choice, according to the BLS, is that inferring a financial relationship among individuals who are not legally related is more uncertain than inferring such a relationship among legal family members.

ment outside childhood should also predict children’s adult outcomes. We find little evidence of such predictive power. In some specifications, we also include maternal unemployment outside the child’s formative years as an additional control, absorbing variation associated with broader maternal unemployment over the life cycle.

Third, we assess residual selection bias using bunching tests developed in [Caetano \(2015\)](#) and [Caetano et al. \(2025\)](#). Because exposure cannot be negative, the zero-exposure bin disproportionately includes mothers with very strong labor-market attachment. If unobserved maternal characteristics account for the baseline patterns, children of zero-exposure mothers should systematically outperform those with minimal positive exposure, producing a discontinuity at zero. We find that children’s wages vary smoothly through zero, with no detectable discontinuity conditional on controls.

3.5 Contemporaneous Family Shocks

A second concern is that maternal unemployment may be correlated with contemporaneous shocks that independently affect children’s outcomes. Child-driven shocks, such as illness or behavioral problems, may cause mothers to leave work, while mother-driven shocks, such as health crises or divorce, may simultaneously raise unemployment risk and reduce children’s well-being.

We partially address this concern by controlling for observable characteristics of the child’s environment that may be correlated with family instability and caregiving demands: maternal age at childbirth, spousal labor supply, family structure and location (number of children, urban/rural residence), and the fraction of years in which fathers and grandparents were present in the household. We also account for local labor market conditions using restricted-use county geocodes (NLSY Geocode data), which, while less granular than the neighborhood-level variation emphasized in [Chetty, Hendren, Kline, and Saez \(2014\)](#); [Chetty, Hendren, and Katz \(2016\)](#); [Chetty, Dobbie, Goldman, Porter, and Yang \(2024\)](#), provide meaningful information about community context. The estimated association between maternal unemployment and children’s adult wages remains stable – and in some cases larger – after adding these factors.

3.6 Measurement Error in Permanent Income

A final concern is measurement error in family permanent income. If unemployment is correlated with true permanent income and our measure of permanent income is noisy, then our results may partly reflect the predictive power of true permanent income that our measure does not capture.¹¹

¹¹Family permanent income is subject to measurement error for two reasons. First, income survey responses are well known to be noisy. Second, we do not observe family income for each year of the child’s childhood. Thus, the

To address this problem, we construct an alternative income measure using a different source of information in the NLSY. Specifically, we use the labor market status array to build a weekly earnings series, defined as usual hours worked times the wage at the primary job, aggregate this to annual earnings, and then construct a permanent income measure as in equation (3). This alternative measure has the advantage of covering the entire childhood period, including years when the mother was not interviewed.

However, this measure is not a perfect substitute for family permanent income: it only includes the mother’s labor earnings and omits other important components such as spouse’s income, asset income, and transfers. Directly controlling for it would therefore bias the interpretation of our regressions toward “maternal labor earnings” rather than “family permanent income.” Instead, we use it as an instrument. By doing so, we exploit the correlation between the alternative and the original measure to correct for attenuation bias, while preserving the broader family income concept. To improve relevance, we interact the instrument with indicators of mothers’ marital status and whether she is the household’s primary earner. These dummies are also included as controls in the main regression, so that only their interactions enter as excluded instruments.

To address measurement error, we estimate equation (1) using two-stage least squares with this instrument. The relevance condition requires maternal permanent labor income to be empirically correlated with broader family permanent income, which we verify in the first stage. The identifying assumption is that, conditional on the controls, the excluded variation in the instrument is related to children’s adult outcomes only through family permanent income. This assumption could fail if maternal labor income also proxies for unobserved maternal ability, labor-market attachment, or other traits that directly shape children’s outcomes. We mitigate this concern by including controls for maternal background and ability proxies, and interpret the IV estimates as a robustness check for measurement error in family permanent income.

4 Results: Baseline Associations and the Anatomy of the Age Gradient

We document the association between exposure to maternal unemployment and children’s adult labor-market outcomes. Our results show that children who were more exposed to maternal unemployment have approximately 4% lower wages as adults. We then unpack this relationship by examining its association with educational attainment, how it varies with the child’s age at exposure, and how mothers adjust time and expenditure inputs during unemployment.

We find strong evidence of age-dependent associations: maternal unemployment exposure

average family income is computed from only a subset of years. On the other hand, our measure of exposure to maternal unemployment is less likely to be subject to measurement error, as it is constructed from the labor market status array. The BLS spends a significant amount of time constructing those and ensuring their quality.

during early childhood has little predictive power for adult outcomes, while exposure during adolescence is a strong predictor of lower wages and employment. For young children, unemployed mothers seem to partially substitute increased time for reduced expenditures on developmental activities, and we find no significant association between early-childhood exposure and measures of cognitive development. By contrast, adolescents exposed to maternal unemployment enter the labor force earlier, consistent with exposure during this period altering educational investments and the school-to-work transition.

4.1 Association with Labor Market Outcomes

We begin by estimating equation (1) to document the baseline association between maternal labor market status and children’s adult outcomes. We examine total earnings, wages, workweek, and employment probability. In Appendix B, we also show results for a measure of occupation risk.¹²

In Table 1, Column 1, we document that children whose mothers spent more time unemployed or out of the labor force during their childhood had lower earnings as adults. The estimated coefficient on the unemployment exposure measure is -0.66 (standard error 0.15) and on the labor-force non-participation exposure measure is -0.32 (standard error 0.04). A child whose exposure to maternal unemployment was one standard deviation above the mean had, on average, 5% lower earnings. In the case of labor-force non-participation exposure, one standard deviation above the mean is associated with 9% lower earnings.

In Columns 2 and 3, we show that the association with earnings comes mainly through wages rather than hours worked. The estimated coefficient of unemployment exposure on wages is -0.50 (standard error 0.09), meaning that one standard deviation higher exposure to maternal unemployment is associated with 4% lower wages. By contrast, the association between unemployment exposure and hours worked is positive but statistically insignificant. We also find that higher exposure to maternal labor-force non-participation is associated with lower wages (-0.16, standard error 0.02) and fewer hours.

We focus on young adults who were employed during the survey week in Columns 1-3, while, in Column 4, we look at the full sample of young adults who reported any employment status. Notice that the sample size is 2.5 times larger in Column 4 than in Columns 1-3. While 70% of those in Column 4 are employed, only half reported wages and hours. We document that those who are more exposed to maternal non-employment are less likely to be employed. The associations with

¹²In Appendix B, Tables B11 and B12 show that greater maternal unemployment exposure during childhood is associated with lower occupational earning risk in adulthood, consistent with individuals sorting into safer but lower-earning occupations. The association is more pronounced for individuals over 30 years old, likely because older workers have had more time to transition into occupations that better match their risk preferences. This finding aligns with Hegarty (2025), who documents a similar pattern using the Panel Study of Income Dynamics (PSID).

Table 1: Association Between Mother’s Labor Market Status and Child’s Outcomes

	(1) log(total earn.)	(2) log(wage)	(3) log(wkly hours)	(4) employed
OLF	-0.321** (0.038)	-0.161** (0.021)	-0.035** (0.009)	-0.226** (0.015)
UNEMP	-0.664** (0.152)	-0.500** (0.085)	0.035 (0.034)	-0.340** (0.065)
Observations	8,687	8,687	8,687	22,920
R2	0.166	0.167	0.059	0.067

Note: OLF and UNEMP are our exposure measures to maternal labor-force non-participation and unemployment, respectively. See equation (2) for construction details. Controls include dummy variables for children’s race and gender, a cubic polynomial in children’s age, and fixed effects for the survey year. All models are estimated by ordinary least squares. Standard errors are clustered at the children’s level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

unemployment and out-of-the-labor-force exposure are similar in magnitude for employment probability. So, while the association with the intensive margin of labor supply is small, the association with the extensive margin is large and significant.

4.2 Association with Educational Attainment

Carneiro and Heckman (2003), Caucutt and Lochner (2020), and others have shown that family income predicts children’s educational attainment. In Table 2, we examine whether maternal unemployment exposure similarly predicts educational attainment, estimating equation (1) with indicators for high school dropout, college attendance, and college graduation as outcomes.

We find that maternal unemployment exposure is a strong predictor of educational attainment across all three margins. In Column 1, a 1 standard deviation increase in unemployment exposure is associated with a 3.8 percentage point increase in the probability of dropping out of high school. In Column 2, the same increase in exposure is associated with a 4.4 percentage point lower probability of attending college, and in Column 3, a 1.7 percentage point lower probability of graduating from college. The coefficients on OLF exposure follow the same pattern but are smaller in magnitude. These associations are all statistically significant at the 5% level.

Since maternal unemployment exposure predicts lower educational attainment, we examine how the documented association with wages and employment changes when conditioning on education. We estimate equation (1) with educational attainment dummies as additional controls.¹³

¹³Because educational attainment is plausibly associated with maternal unemployment exposure, specifications including education controls should be interpreted as conditional associations within education groups rather than total associations.

Table 2: Association with Educational Attainment

	(1) HS dropout (ages 21–24)	(2) Attended college (ages 24–27)	(3) Graduated college (ages 24–27)
OLF	0.186** (0.033)	-0.206** (0.020)	-0.073** (0.012)
UNEMP	0.504** (0.124)	-0.580** (0.070)	-0.231** (0.045)
Observations	3,516	7,719	7,719
R2	0.293	0.242	0.359

Note: The dependent variables are indicator variables constructed from the respondent’s highest grade completed as of the survey date. To align observed educational attainment with likely schooling completion, the Attended College and Graduated College indicators are defined for ages 24 to 27, while the Dropout indicator is defined for ages 21 to 24. OLF and UNEMP are our exposure measures to maternal labor-force non-participation and unemployment, respectively. See equation (2) for construction details. Controls include dummy variables for children’s race and gender, a cubic polynomial in children’s age, and fixed effects for the survey year. All models are estimated by ordinary least squares. Standard errors are clustered at the children’s level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

Table 3 presents the results. Columns 1 and 3 reproduce the baseline associations for wages and employment, while Columns 2 and 4 add education controls.

After conditioning on educational attainment, the coefficient on unemployment exposure decreases from -0.502 to -0.359 for wages (a 28% reduction) and from -0.386 to -0.264 for employment probability (a 32% reduction). The education controls themselves are strongly associated with outcomes: relative to high school dropouts, college graduates earn approximately 39 log points more and have a 26 percentage point higher employment probability. These patterns confirm that education is an important channel through which maternal unemployment exposure predicts adult outcomes. At the same time, the associations remain negative and statistically significant after conditioning on education, indicating that maternal unemployment exposure retains predictive power for adult wages and employment even within education groups.

4.3 The Role of Timing: Exposure Across Childhood Stages

We now examine whether the association between maternal unemployment and children’s adult outcomes varies with the child’s age at exposure. To do this, we estimate a modified version of Equation (1), allowing the measures of exposure to maternal unemployment and non-participation to differ across developmental stages:

$$y_{it} = \alpha + \sum_j \beta_{1,j} UNEMP_{i,j} + \sum_j \beta_{2,j} OLF_{i,j} + \gamma_1 X_i + \gamma_2 Z_{it} + \epsilon_{it} . \quad (4)$$

Table 3: Association After Conditioning on Educational Attainment

	(1)	(2)	(3)	(4)
	log(wage)	log(wage)	emp.	emp.
OLF	-0.178** (0.024)	-0.113** (0.022)	-0.205** (0.017)	-0.150** (0.017)
UNEMP	-0.502** (0.089)	-0.359** (0.079)	-0.386** (0.067)	-0.264** (0.065)
High School		0.030** (0.015)		0.078** (0.012)
Some College		0.189** (0.017)		0.179** (0.012)
College or more		0.391** (0.020)		0.259** (0.013)
Observations	8,021	8,021	20,857	20,857
R2	0.162	0.247	0.059	0.088

Note: OLF and UNEMP are our exposure measures to maternal labor-force non-participation and unemployment, respectively. See equation (2) for construction details. Baseline controls include dummy variables for children's race and gender, a cubic polynomial in children's age, and fixed effects for the survey year. Columns 2 and 4 additionally include dummy variables for children's educational attainment. All models are estimated by ordinary least squares. Standard errors are clustered at the children's level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

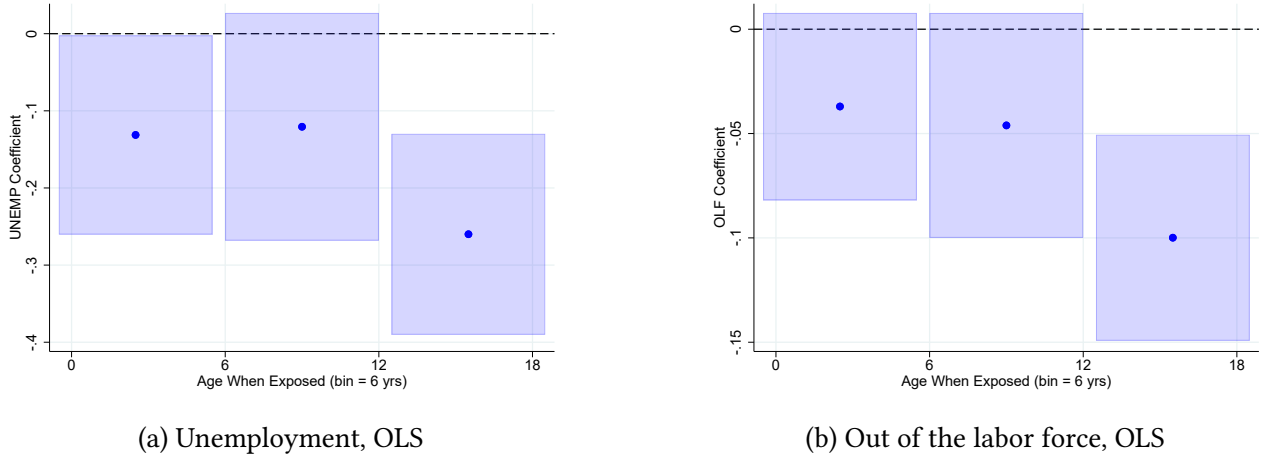
We use 6-year bins for the development stages to smooth the results, grouping them as 0–5, 6–11, and 12–17.¹⁴

Figure 2 presents the results. Panel (a) shows the association between maternal unemployment exposure and children's adult wages by age of exposure. The estimated associations are concentrated in adolescence. Exposure during early childhood (ages 0–5) and mid-childhood (ages 6–11) has negative but statistically insignificant coefficients. By contrast, exposure during adolescence (ages 12–17) is a statistically significant predictor of lower adult wages. Panel (b) shows the corresponding pattern for maternal labor-force non-participation. OLF exposure during adolescence is also associated with lower wages, while the earlier-age coefficients are imprecisely estimated.

Appendix C.1 shows that the age gradient is robust to conditioning on permanent income and to correcting for measurement error in permanent income using our IV strategy. The pattern is unchanged: the predictive power of maternal unemployment for adult wages is concentrated in adolescence.

¹⁴We restrict analysis to ages 0–18 and exclude pre-birth unemployment. While pre-birth unemployment serves as a useful placebo test (Section 5.3), including it in the life-cycle specification reduces the sample from 7,672 to 4,164 observations (Table 9) because many mothers are not yet active in the labor force before childbirth. This sample restriction creates composition effects that confound the interpretation of age-specific patterns.

Figure 2: Association Between Maternal Employment Status and Wages by Childhood Stage



Note: This figure plots the estimated associations between maternal unemployment (panel a) and labor-force non-participation (panel b) and children's log wages, by age of exposure. All specifications control for child demographics (race, sex, age polynomials), and fixed effects for the survey year. The x-axis represents the child's age at exposure, grouped into 6-year bins. Point estimates are shown with 95% confidence intervals, with standard errors clustered at the individual level.

4.4 Early Childhood: Parental Time, Expenditure, and Child Development

The age gradient documented above shows that early childhood exposure to maternal unemployment (ages 0-5) has little predictive power for adult wages. We now examine patterns that may help explain this finding. We use the ATUS and CE to document how mothers' time allocation and household expenditures differ by labor market status, and then examine whether early-childhood exposure is associated with measures of the home environment and cognitive development in the NLSY-CYA.

Time Allocation by Mothers' Labor Market Status

We use the American Time Use Survey (ATUS) to document how mothers' time allocation varies with their labor market status. The ATUS measures how individuals divide their time among various activities. We focus on women between 25 and 50 who reported having children under 18 living in the household. Appendix A provides a detailed description of the ATUS data and our sample.

In Table 4, we compare the time allocation of these women when they are employed, unemployed, and out of the labor force. We compute the mean time spent on different activities and reweight the observations to correct for differences in the number of children across women with different employment statuses. The reweighting factor is the inverse of each employment status conditional probability on the number of children, discretized into four bins.

The largest difference between employed and unemployed mothers is in work and work-related activities: an employed mother works, on average, 6.5 hours, while an unemployed mother works only 30 minutes. Unemployed mothers distribute this extra time across several activities. They increase the time allocated to personal care and household activities by 137 minutes and to leisure by 105 minutes. Crucially, they also increase time allocated to activities related to children's education by 8 minutes – a small absolute increase, but one that represents a 77% increase relative to employed mothers. Unemployed mothers thus nearly double the time devoted to children's educational activities.

Mothers out of the labor force spend even more time with their children than unemployed mothers. This pattern is consistent with selection: mothers who choose to be out of the labor force may place a higher value on time with children, which could also explain why the associations between OLF exposure and children's adult outcomes are weaker than those for unemployment exposure.

In Tables B13, B14, and B15, we reweight to correct for other observable differences across labor market status groups, including family income, education levels, and the age of the youngest child. The time allocated to children's education is consistently higher for unemployed mothers than for employed mothers, by 70% to 80%. This pattern is robust across all reweighting schemes.

Expenditure Patterns by Mothers' Labor Market Status

Unemployed mothers nearly double the time allocated to children's educational activities (Table 4). We now investigate whether this increase in home-based time is accompanied by lower spending on market-based inputs. We use the Consumer Expenditure Survey (CE) to document the association between parental labor market status and child-related expenditures. Appendix A provides a detailed description of the CE data and our sample.

The CE provides information on monthly household expenditures, family structure, and employment statuses of each household member. We aggregate expenditures into quarters and identify child-related expenditures based on expenditure category names in the CE, including childcare, school and college tuition, apparel, and entertainment. Appendix A provides a complete description of these categories. We classify non-working parents as unemployed if they (i) are not working and (ii-a) report inability to find a job as the reason for not working or (ii-b) have received unemployment benefits over the past 12 months. All other non-working parents are classified as out of the labor force.

We estimate OLS regressions with two dependent variables: log total child-related expenditures (Columns 1-2) and the share of expenditures devoted to childcare (Columns 3-4). The main

Table 4: Average Number of Minutes by Mother's Labor Market Status

	Emp	Unemp	OLF
Total Time Spent on All Activities	1440.00	1440.00	1440.00
Caring for and Helping Household Members	112.44	155.89	193.77
Caring for and Helping Children	90.83	132.75	169.39
Education-Related Activities for Children	10.74	19.00	21.10
Health-Related Activities for Children	3.17	4.77	4.79
Other Caring Activities for Children	76.92	108.97	143.50
Caring for and Helping Non-Household Members	6.40	14.47	11.92
Working and Work-Related Activities	384.05	35.84	5.25
Leisure and Social Activities	172.24	276.96	266.43
Purchasing Goods and Eating	103.09	125.69	130.45
Personal Care and Household Activities	634.48	771.85	778.23
Educational Activities	7.25	27.58	20.38
Other and Communication Activities	20.05	31.72	33.57

Note: The sample consists of women between 25 and 50 who reported having children under 18 living in the household. We look at the days of the week and non-holidays. We have 14,755 employed, 1,014 unemployed, and 5,135 out-of-the-labor-force respondents. Time allocation categories are the major BLS aggregate categories. Each cell represents averages. We use survey sample weights and construct additional weights to correct for differences in the number of children across employment status groups.

variables of interest are the labor market statuses of both parents. Controls include total expenditures, family size, the number of children younger than 16 years old, parental education, and time fixed effects. Households with no children under 16 are excluded.

Table 5, Column (1) shows that households with non-employed mothers spend approximately 30% less on child-related expenditures relative to families with employed mothers. Households with non-employed fathers also spend less, though the magnitude is smaller. Column (2) adds controls for parental education, which reduces the associations. For mothers, the coefficients are larger for OLF than for unemployment, which could reflect unemployed mothers spending time on job search and therefore not fully substituting for market-based inputs, or unobserved differences between the two groups. For fathers, the pattern is reversed: unemployment is associated with slightly larger reductions than OLF, though these differences are small.

Columns (3) and (4) examine the share of expenditures dedicated to childcare among households with children under two. The childcare share is, on average, 6-7 percentage points lower in households with non-working mothers relative to those with employed mothers. The association is stronger for OLF mothers, consistent with Columns (1) and (2). Paternal employment status has a much weaker association, suggesting that fathers do not tend to substitute for market-based childcare. Together with the time-use evidence, these results are consistent with a pattern of input substitution: unemployed mothers increase home-based time with children while reducing market-based expenditures.

Table 5: Child-Related Expenditures by Parental Labor Market Status (CE)

	(1)	(2)	(3)	(4)
	log kids exp	log kids exp	childcare share	childcare share
Mother OLF	-0.339** (0.012)	-0.314** (0.011)	-0.075** (0.003)	-0.072** (0.003)
Mother Unemployed	-0.282** (0.056)	-0.246** (0.056)	-0.060** (0.013)	-0.056** (0.012)
Father OLF	-0.166** (0.025)	-0.148** (0.024)	-0.014** (0.007)	-0.015** (0.007)
Father Unemployed	-0.192** (0.063)	-0.171** (0.062)	-0.036** (0.015)	-0.029** (0.015)
Constant	-0.658** (0.068)	0.001 (0.174)	0.087** (0.017)	0.175** (0.038)
Number of children	Yes	Yes	Yes	Yes
Family size	Yes	Yes	Yes	Yes
Total expenditures	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Parental education		Yes		Yes
Observations	61,424	61,424	13,636	13,636
R2	0.220	0.233	0.093	0.111

Note: Mother OLF/Unemployed and Father OLF/Unemployed represent dummy variables corresponding to labor statuses of each parent. Base categories are working mother and working father. Baseline controls include the number of children younger than 16 years old, the number of household members, total expenditures and year- and quarter-fixed effects. Columns (2) and (4) additionally account for parental education, defined as the maximum of the maternal and paternal educational levels. Sample size in Columns (3) and (4) is smaller, since it includes only families with children under the age of 2. All models are estimated by ordinary least squares. Standard errors are reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

Association with Home Environment and Child Development

Association with Home Environment and Child Development

We now examine whether early-childhood maternal unemployment exposure is associated with measures of cognitive development and home environment quality. We use two assessments from the NLSY-CYA. The Peabody Individual Achievement Test (PIAT) measures cognitive achievement. The HOME (Home Observation for Measurement of the Environment) Inventory measures the quality of the child's home environment, including emotional support and cognitive stimulation.

In Table 6, we regress standardized scores on maternal unemployment and out-of-labor-force exposure during ages 0-5, with scores measured between ages 7 and 9. Controls include maternal education and AFQT quintile dummies, child race and gender dummies, a cubic polynomial in child age, and survey-year fixed effects.

The PIAT results provide little evidence that early-childhood unemployment exposure predicts lower cognitive achievement by ages 7-9. This pattern is consistent with the time-use and expenditure evidence documented above: if unemployed mothers partially substitute additional time for reduced market inputs, the net effect on measured cognitive development may be limited.

Table 6: Association with Home Environment and Child Development

	(1)	(2)	(3)	(4)	(5)
	HOME Total	Emotional	Cognitive	Avg. PIAT	Math PIAT
OLF 0-5	-0.199** (0.049)	-0.125** (0.056)	-0.189** (0.053)	-0.074 (0.053)	-0.075 (0.053)
UNEMP 0-5	-0.743** (0.195)	-0.859** (0.209)	-0.405* (0.212)	-0.177 (0.183)	0.054 (0.189)
Mom Educ Dummies	Yes	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes	Yes
Observations	6,863	6,532	6,701	6,619	6,619
R2	0.299	0.203	0.244	0.185	0.180

Note: OLF and UNEMP are our exposure measures to maternal labor-force non-participation and unemployment, respectively, measured during ages 0-5. See equation (2) for construction details. Controls include maternal education attainment and AFQT quintile dummies, dummy variables for children's race and gender, a cubic polynomial in children's age, and survey-year fixed effects. Dependent variables are total HOME score, emotional support HOME score, cognitive stimulation HOME score, average PIAT score, and PIAT Math score, all measured between ages 7 and 9. The average PIAT score is an average of the PIAT Math, PIAT Reading Recognition, and Reading Comprehension scores. All models are estimated by ordinary least squares. Standard errors are clustered at the children's level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

The picture is different for the home environment. The OLS estimates reveal negative, statistically significant associations between early-childhood unemployment exposure and HOME scores. A one-unit increase in exposure is associated with a 0.743 standard deviation reduction in the overall HOME score (standard error 0.195) and a 0.859 standard deviation reduction in the Emotional subscale (standard error 0.209). The Cognitive subscale association is also negative (-0.405, standard error 0.212). These associations likely reflect the broader disruption that unemployment brings to household dynamics.

The HOME associations raise the question of what channel they reflect. In Appendix Table B3, we report IV estimates that condition on permanent income, correcting for measurement error as described in Subsection 3.6. Conditioning on permanent income substantially attenuates the UNEMP 0-5 coefficients, suggesting that a meaningful share of the OLS association operates through the income channel. The IV estimates are imprecise, however, with confidence intervals that include both zero and economically meaningful negative associations, so we do not interpret them as establishing the absence of a relationship.

The comparison between OLS and IV suggests that measured family resources are relevant for interpreting the association between early-childhood unemployment exposure and home environ-

ment quality. The point estimates attenuate after conditioning on permanent income, although the IV estimates are imprecise. This attenuation pattern contrasts with adolescent exposure, where the association with adult wages persists after conditioning on permanent income (Section 5). Taken together – limited evidence of lower PIAT scores, input substitution in time use and expenditures, and attenuation of the HOME associations after conditioning on permanent income – these patterns are consistent with the interpretation that early-childhood unemployment operates partly through family resources rather than through lasting disruptions to cognitive development.

4.5 Late Childhood: School-to-Work Transition

The age gradient shows that maternal unemployment exposure during adolescence (ages 12-17) is a strong predictor of lower adult wages, while early-childhood exposure has little predictive power. We now examine a pattern that may help explain this concentration: the association between adolescent exposure and the timing of labor market entry. Adolescence coincides with educational and career decisions, and earlier workforce entry is associated with lower educational attainment and potentially worse initial job matches.

To measure the timing of labor market entry, we use NLSY-CYA work history records containing the calendar month and year each job began. We define labor market entry as the age at which the respondent first holds a full-time position (average weekly hours exceeding 35), restricting attention to jobs starting at age 14 or later to exclude casual youth employment. This definition captures meaningful workforce entry rather than temporary or part-time work during school.

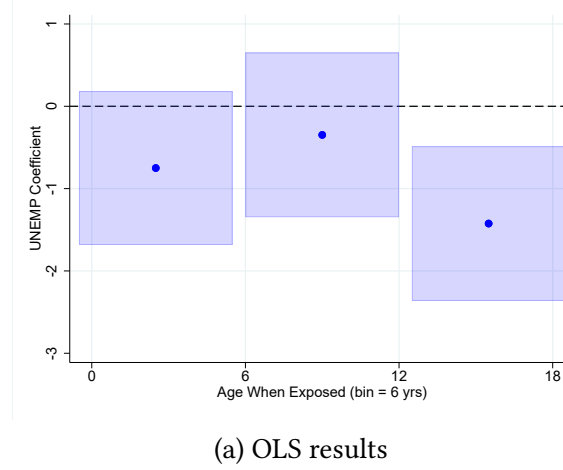
Figure 3 presents OLS estimates of the association between maternal unemployment exposure and entry age by childhood stage. Exposure during adolescence (ages 12-17) is associated with significantly earlier full-time labor-market entry. By contrast, exposure during early childhood shows no significant association with entry timing. This pattern mirrors the age gradient for adult wages and is consistent with Fradkin et al. (2019), who document that children of displaced workers in Belgium enter the labor market earlier. Appendix C.1 shows that the results are robust to conditioning on permanent income and correcting for measurement error using our IV strategy.

Earlier labor market entry is consistent with lower educational investment. Recall that Table 2 documents that maternal unemployment exposure is associated with higher high school dropout rates and lower college attendance and graduation. Together with the association with earlier workforce entry, these patterns suggest that adolescent exposure to maternal unemployment is linked to a simultaneous acceleration of labor market entry and reduction in schooling.

Exposure to maternal unemployment may also be associated with occupational sorting. Following Bonin, Dohmen, Falk, Huffman, and Sunde (2007), Hartog and Vijverberg (2007), and Necker

and Voskort (2014), we construct measures of occupational earnings risk. Table B11 shows that greater unemployment exposure is associated with employment in occupations with lower earnings risk. This association is stronger among workers over age 30, consistent with gradual sorting into lower-risk career paths. This pattern aligns with Hegarty (2025), who document a similar relationship between parental layoffs and occupational risk using PSID data.

Figure 3: Association Between Maternal Unemployment and Age at Labor Force Entry



Note: This figure plots the estimated association between maternal unemployment exposure and children's age at labor market entry, by age of exposure. Labor market entry is defined as the age at which the respondent first holds a full-time position (average weekly hours exceeding 35 hours), restricting attention to jobs starting at age 14 or later. Controls include child demographics (race, sex, age polynomials) and fixed effects for the survey year. The x-axis represents the child's age at exposure, grouped into 6-year bins. Point estimates are shown with 95% confidence intervals, with standard errors clustered at the individual level.

5 Conditioning on Family Resources and Selection Diagnostics

The previous section documented the baseline association between maternal unemployment exposure and children's adult outcomes, its connection to educational attainment, and the age gradient concentrating the predictive power in adolescence. We now investigate the extent to which these associations persist after conditioning on family permanent income, maternal ability proxies, and other observable characteristics. We also implement diagnostic tests – placebo regressions using out-of-window exposure and bunching tests at zero exposure – to assess whether selection on unobserved maternal traits can account for the baseline patterns. Throughout this section, we condition on family permanent income and correct for measurement error using the IV strategy described in Subsection 3.6.

5.1 Conditioning on Family Permanent Income

A central question is the extent to which the association between maternal unemployment and children’s adult outcomes reflects lost family income versus other channels. To assess this, we augment equation (1) by conditioning on family permanent income, constructed following [Carneiro and Heckman \(2003\)](#) as described in Section 3.

Table 7, Columns 1 and 4 reproduce the baseline OLS results for wages and employment probability, respectively. Columns 2 and 5 add permanent income as a control. The association between maternal unemployment exposure and children’s outcomes shrinks but persists: conditional on permanent income, one standard deviation higher exposure is still associated with approximately 3% lower wages. The coefficients decrease between 9% and 31% relative to the baseline, suggesting that income loss accounts for a meaningful but incomplete share of the overall association.

We address the possibility of measurement error in permanent income using the two-stage least squares strategy described in Subsection 3.6. Table 7, Column 3 shows that the coefficient on permanent income increases from 0.099 to 0.136 when corrected for attenuation bias, implying that each \$10,000 increase in average family income during childhood is associated with 0.136 log points higher adult wages. Column 6 shows the corresponding result for employment probability, with each \$10,000 increase associated with a 2.2 percentage point higher probability of employment. After correcting for measurement error, the estimated associations with maternal unemployment and labor-force non-participation decrease relative to the OLS results but remain negative and statistically significant.

To put these magnitudes in perspective, one standard deviation more exposure to maternal unemployment is associated with a wage reduction similar to that associated with \$1,645 lower average family income during childhood.¹⁵

These results indicate that while family income is an important predictor of children’s adult outcomes – consistent with [Carneiro and Heckman \(2003\)](#) and [Caucutt and Lochner \(2020\)](#) – maternal unemployment exposure retains substantial predictive power after conditioning on permanent income. This suggests that non-income aspects of unemployment, such as disruptions to household routines, parental stress, or changes in children’s own behavior, are independently associated with worse adult outcomes.

¹⁵Appendix C.4 shows that an implausibly large attenuation bias in the permanent income coefficient would be required to eliminate the association with unemployment exposure.

Table 7: Association After Conditioning on Family Permanent Income

	OLS		IV	OLS		IV
	(1)	(2)	(3)	(4)	(5)	(6)
	log(wage)	log(wage)	log(wage)	emp.	emp.	emp.
OLF	-0.176** (0.023)	-0.123** (0.024)	-0.103** (0.026)	-0.201** (0.017)	-0.184** (0.017)	-0.189** (0.018)
UNEMP	-0.393** (0.085)	-0.329** (0.077)	-0.305** (0.078)	-0.303** (0.068)	-0.275** (0.067)	-0.284** (0.068)
Permanent Income		0.099** (0.010)	0.137** (0.021)		0.032** (0.006)	0.022* (0.012)
F statistic			59.594			104.215
Observations	8,382	8,382	8,382	21,737	21,737	21,737
R2	0.175	0.196	0.193	0.065	0.067	0.066

Note: OLF and UNEMP are our exposure measures to maternal labor-force non-participation and unemployment, respectively. See equation (2) for construction details. Permanent Income is in \$10,000s (discounted present value) as of birth year. See equation (3) for construction details. Controls include dummy variables for children's race and gender, a cubic polynomial in children's age, dummy variables for mothers' marital and primary earner statuses, and fixed effects for the survey year. Columns 1, 2, 4, and 5 are estimated by ordinary least squares. Columns 3 and 6 are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses. The F statistic is the Kleibergen-Paap F statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

5.2 Selection on Mothers' Unobserved Heterogeneity: Ability Proxies

The association between maternal unemployment and children's outcomes may partly reflect unobserved maternal heterogeneity, such as differences in the ability to invest in children's human capital. In Table 8, we progressively add three proxies that capture maternal abilities: (1) mothers' educational attainment, (2) cognitive test scores (AFQT), and (3) time spent unemployed outside the first 18 years of the child's life. Columns 2-4 gradually introduce these proxies, while Column 1 reports the results without any proxy.

Looking across Columns 2 to 4, the ability proxies account for only a modest share of the association between unemployment exposure and adult wages. With all proxies included (Column 4), one standard deviation higher exposure remains associated with approximately 1.5% lower wages. The proxies account for a larger share of the association with OLF exposure, making it statistically insignificant – consistent with OLF status being more closely tied to unobserved maternal characteristics than to unemployment. Column 5 shows a similar pattern for employment probability: the ability proxies explain only a small part of the association with unemployment exposure.

Table 8: Measures of Mother's Ability: Effects on Wages and Employment

	(1)	(2)	(3)	(4)	(5)
	log(wage)	log(wage)	log(wage)	log(wage)	emp.
OLF	-0.055** (0.025)	-0.041 (0.026)	-0.036 (0.026)	-0.035 (0.026)	-0.132** (0.019)
UNEMP	-0.266** (0.073)	-0.252** (0.073)	-0.229** (0.073)	-0.212** (0.078)	-0.137* (0.070)
Permanent Income	0.115** (0.022)	0.107** (0.024)	0.102** (0.024)	0.102** (0.024)	0.008 (0.014)
Educ Dummies	Yes	Yes	Yes	Yes	Yes
Mom Educ Dummies		Yes	Yes	Yes	Yes
Mom AFQT			Yes	Yes	Yes
Remaining Unemp				Yes	Yes
F statistic	58.868	51.283	49.798	49.977	77.825
Observations	7,710	7,710	7,710	7,710	20,102
R2	0.268	0.270	0.273	0.273	0.094

Note: OLF and UNEMP are our exposure measures to maternal labor-force non-participation and unemployment, respectively. See equation (2) for construction details. Permanent Income is in \$10,000s (discounted present value) as of birth year. See equation (3) for construction details. Column 1 only has the baseline controls, while other columns have additional controls. Column 2 includes mother's education attainment dummies. Column 3 additionally controls for mothers' Armed Forces Qualification Test (AFQT) scores. We include the test scores as quantiles of the score distribution. Column 4 also includes the mother's time spent unemployed between ages 25-60, excluding the first 18 years of the child's life. This control captures unobservable characteristics that cannot be captured by education achievements or cognitive test scores. Baseline controls include dummy variables for children's race, gender, and educational attainment, a cubic polynomial in children's age, dummy variables for mothers' marital and primary earner statuses, and fixed effects for the survey year. All models are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses. The F statistic is the Kleibergen-Paap F statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

Birth and Prenatal Characteristics. We also examine whether birth and prenatal characteristics account for the baseline associations. Specifically, we control for the child's birth order, gestation length (in weeks), birth weight (in ounces), and the shortest birth spacing among siblings. The last control is of particular interest, given evidence from [Dougan et al. \(2025\)](#) on the importance of birth spacing for the effectiveness of early childhood interventions. Table B9 shows that these controls do not meaningfully change the estimated association between maternal unemployment exposure and children's adult outcomes.

5.3 Placebo Tests Using Out-of-Window Exposure

We conduct a placebo test by replacing our main explanatory variable – maternal unemployment during ages 0-18 – with maternal unemployment outside that period (before the child’s birth and after age 25). The logic is straightforward: if the baseline association reflects persistent maternal traits (low ability, poor health) that independently predict children’s outcomes, then unemployment at any point in the mother’s career should have similar predictive power. Finding that only exposure during childhood matters would be consistent with the timing of exposure being relevant, rather than the association being driven entirely by selection on persistent maternal characteristics.

The results are presented in Table 9. We include unemployment exposure when children are already adults (ages 25-30) and before they are born (5 years to 1 month before birth). Columns 1-3 show that maternal unemployment outside the child’s formative years has no statistically significant association with adult wages. This contrasts with our main results, where maternal unemployment during ages 0-18 is a strong predictor of lower wages. Column 4 shows that out-of-window unemployment similarly has no association with employment probability.

5.4 Bunching Tests at Zero Exposure

We employ a bunching test at the point of zero unemployment (Caetano, 2015; Caetano et al., 2025). Unemployment exposure is truncated at zero: mothers cannot have negative exposure. This mechanical constraint creates a composition issue at the boundary. The zero-exposure bin disproportionately includes mothers with very strong labor-market attachment – mothers who, absent the truncation, would have had “negative unemployment.” As a result, the zero-exposure group has a higher concentration of high-ability, stable-employment mothers compared to the group with small positive exposure.

This composition difference has testable implications. If unobserved maternal characteristics (ability, labor-market attachment) account for the baseline associations, we should observe a discontinuity at zero: children of zero-exposure mothers would, on average, have better outcomes than children whose mothers had minimal positive exposure. By contrast, if the association reflects exposure itself, outcomes should vary smoothly through zero, since a mother with zero exposure and a mother with 0.01 years of exposure differ negligibly in actual unemployment.

We implement this test graphically using local linear regressions for two outcomes: children’s wages and maternal grandfather’s education. Grandfather’s education serves as a placebo outcome that should not be associated with maternal unemployment, conditional on observables. Figure 4, Panels (a) and (b) show that, without controls, both outcomes display discontinuities at zero unemployment, raising initial concerns about selection.

Table 9: Placebo test

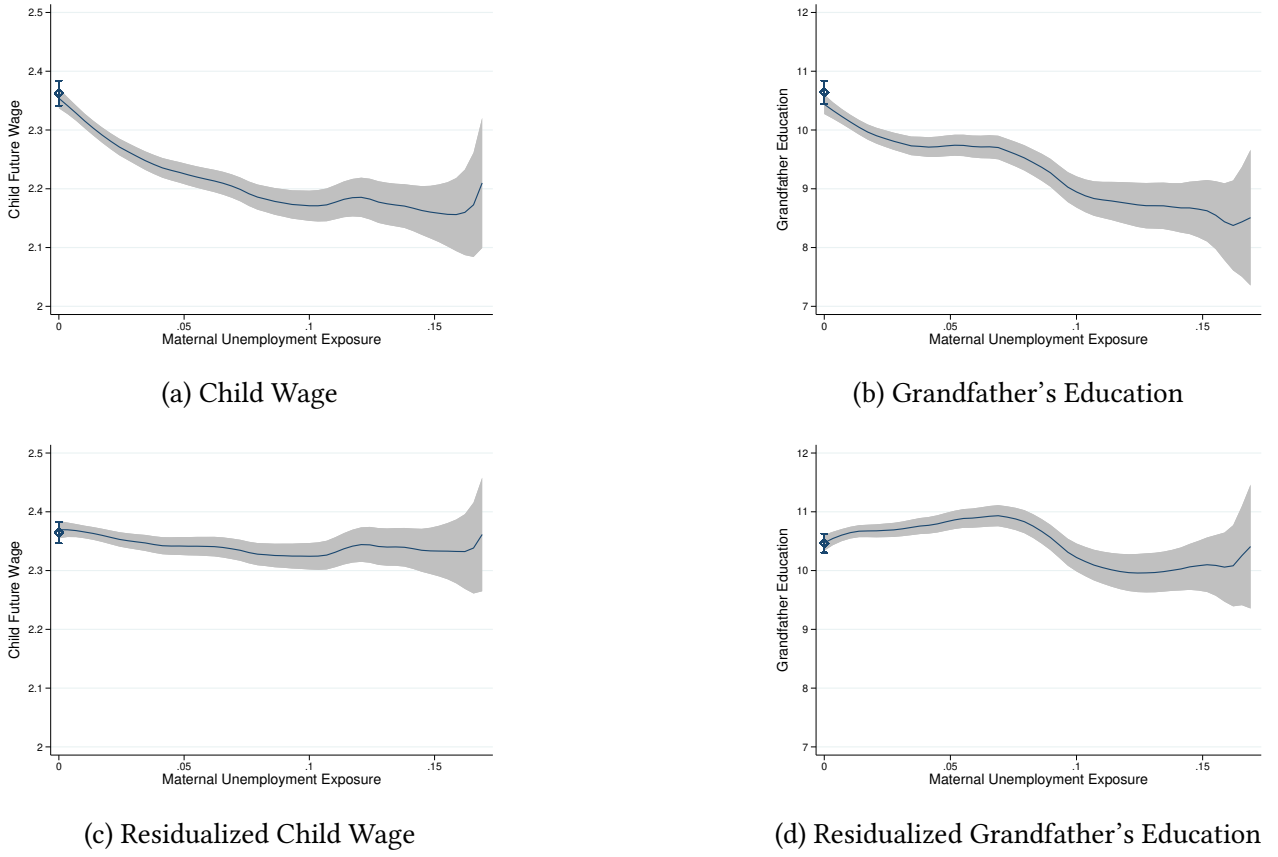
	(1)	(2)	(3)	(4)
	log(wage)	log(wage)	log(wage)	emp.
OLF when child is bw 25-30	-0.024 (0.018)		-0.005 (0.023)	-0.052** (0.017)
OLF when child is bw -5-0		-0.042 (0.026)	-0.044* (0.027)	-0.006 (0.021)
UNEMP when child is bw 25-30	-0.056 (0.051)		-0.042 (0.066)	-0.091 (0.060)
UNEMP when child is bw -5-0		0.016 (0.072)	0.015 (0.073)	-0.033 (0.053)
Permanent Income	0.120** (0.023)	0.058** (0.024)	0.058** (0.023)	0.044** (0.016)
Educ Dummies	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes
Mom Controls	Yes	Yes	Yes	Yes
F statistic	81.338	48.639	49.460	75.737
Observations	7,672	4,164	4,123	11,489
R2	0.269	0.291	0.291	0.102

Note: OLF and UNEMP are our exposure measures to maternal labor-force non-participation and unemployment, respectively. See equation (2) for construction details. Permanent Income is in \$10,000s (discounted present value) as of birth year. See equation (3) for construction details. Controls include dummy variables for children's race, gender, and educational attainment, a cubic polynomial in children's age, dummy variables for mothers' marital and primary earner statuses, mothers' ability controls, and fixed effects for the survey year. For mothers' ability, we use mothers' education attainment dummies and mothers' Armed Forces Qualification Test (AFQT) scores, which we include as quantiles. We do not include the mother's time spent unemployed between ages 25-60, excluding the first 18 years of the child's life. This variable is extremely correlated with our placebo measures. All models are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses. The F statistic is the Kleibergen-Paap F statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

However, Panels (c) and (d) show that after including proxies for maternal ability and other controls, no discontinuities remain at the zero-unemployment threshold. After conditioning on observables, maternal unemployment exposure is smoothly and negatively associated with children's wages across the full range of exposure. By contrast, the relationship between grandfather's education and maternal unemployment exposure changes sign: it is positive for exposures up to 1.8 years (21 months) and turns slightly negative thereafter. Together, this evidence reduces concerns that the baseline associations are driven by selection.

Figure 4: Bunching Test: Outcome and Placebo Outcome



Note: This figure presents the bunching test results for both the primary outcome variable (Child Wage) and a placebo outcome (Grandfather's Education). Panels (a) and (b) show raw outcomes, while Panels (c) and (d) display residualized outcomes, controlling for relevant covariates. To compute residualized outcomes, we regress the relevant variable on maternal unemployment exposure and a full set of controls, including child race, sex, age, year-fixed effects, education dummies, and our maternal ability proxies. We add back the variation accounted for maternal unemployment exposure using the estimated coefficient. Lastly, we rescale the estimates of Panels (c) and (d) by adding the mean at zero unemployment computed in Panels (a) and (b). The local polynomial smoothing uses a bandwidth of 0.015. Bootstrapped standard errors are computed using 1,000 replications.

The fact that grandfather's education remains negatively associated with high levels of maternal unemployment exposure in Panel (d) raises the question of whether the results are sensitive to observations in the upper tail of the exposure distribution. In Table 10, we re-estimate our baseline model after restricting the support of unemployment exposure. Column 1 shows the full sample, while Columns 2, 3, and 4 restrict the sample to observations below the 95th, 90th, and 85th percentiles, respectively. The coefficient on unemployment exposure remains negative and becomes more negative as the upper tail is trimmed. Thus, the negative association is not limited to the highest-exposure observations; it is also present over more moderate ranges of maternal unemployment exposure.¹⁶

¹⁶This pattern is broadly consistent with the local polynomial evidence above and with the quartile estimates in Appendix B.2, where the negative association between unemployment exposure and wages becomes larger across quartiles rather than appearing only at the top of the distribution.

In Table 10, we also conduct a regression-based diagnostic for bunching by including a dummy variable indicating zero unemployment exposure. In a specification that also controls linearly for unemployment exposure, this dummy tests whether the zero-exposure group differs from the outcome level implied by the exposure-outcome relationship. We find that the coefficient is statistically indistinguishable from zero, consistent with the graphical evidence of no visible break at the zero-exposure threshold. Following the intuition in Caetano (2015) and Caetano et al. (2025), the test is informative about selection associated with the bunching point: if mothers with zero unemployment exposure were selected on unobserved labor-market attachment, children's outcomes might shift at zero relative to the relationship estimated for positive exposure. The null finding reduces concerns about this specific form of selection. The result holds when excluding tail events (Columns 2-4).

Table 10: Restricting Tail Events: Effects on Wages

	All Sample	UNEMP < p(95)	UNEMP < p(90)	UNEMP < p(85)
	(1)	(2)	(3)	(4)
	log(wage)	log(wage)	log(wage)	log(wage)
$1\{\text{UNEMP} = 0\}$	-0.005 (0.016)	-0.011 (0.017)	-0.013 (0.017)	-0.018 (0.017)
OLF	-0.036 (0.026)	-0.028 (0.026)	-0.030 (0.027)	-0.028 (0.027)
UNEMP	-0.220** (0.080)	-0.394** (0.152)	-0.376* (0.213)	-0.636** (0.252)
Permanent Income	0.103** (0.025)	0.103** (0.026)	0.104** (0.027)	0.103** (0.027)
Educ Dummies	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes
Remaining Unemp	Yes	Yes	Yes	Yes
F statistic	46.297	47.182	45.784	43.485
Observations	7,713	7,328	6,938	6,557
R2	0.273	0.268	0.265	0.268

Note: See Table 8 for the definition of variables and mothers' ability control. All models are estimated by two-stage least squares. Excluded instruments used in all Columns are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses. The F statistic is the Kleibergen-Paap F statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

5.5 Effects Across Labor Market Outcomes

Most of our previous analysis only focuses on wages and employment probability. In this last subsection, we analyze other labor market outcomes using the specification in Column 4 of Table 8. We analyze: (1) total earnings, (2) wages, (3) weekly hours, and (4) employment probability.¹⁷

Column 1 of Table 11 shows that total earnings are negatively associated with OLF exposure (statistically significant) but not with unemployment exposure. This appears to reflect offsetting associations: Column 2 shows that wages decline with unemployment exposure, while Column 3 shows that hours worked increase. These opposing patterns result in a statistically insignificant association with total earnings. Permanent income is positively and significantly associated with both total earnings and wages but not with hours. Column 4 confirms the negative associations of both exposure measures with employment probability.

Table 11: All Labor Market Outcomes

	(1) log(total earn.)	(2) log(wage)	(3) log(hours)	(4) employed
OLF	-0.152** (0.046)	-0.035 (0.026)	-0.031** (0.012)	-0.132** (0.019)
UNEMP	-0.171 (0.150)	-0.213** (0.078)	0.086** (0.037)	-0.137* (0.070)
Permanent Income	0.086** (0.038)	0.103** (0.024)	-0.006 (0.009)	0.008 (0.014)
Educ Dummies	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes
Add. unemp	Yes	Yes	Yes	Yes
F statistic	50.051	50.051	50.051	77.825
Observations	7,713	7,713	7,713	20,102
R2	0.232	0.273	0.069	0.094

Note: See Table 8 for the definition of variables and mothers' ability control. All models are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses. The F statistic is the Kleibergen-Paap F statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

¹⁷This specification contrasts with Table 1 by adding additional controls and addressing measurement error.

5.6 Additional Robustness and Sensitivity Analyses

We conduct a series of robustness checks and heterogeneity analyses to assess the stability of our main results.

Functional Form and Specification Checks. Table B2 tests for non-linear associations between unemployment exposure and adult outcomes and finds no strong departures from linearity. Table B5 decomposes exposure into the number and length of unemployment spells, showing that the frequency of maternal unemployment spells is negatively associated with children's future wages, while average duration is imprecisely estimated.

The panel structure of the NLSY allows us to exploit variation between siblings who experienced different amounts of maternal unemployment. Table B6 reports estimates from models with mother fixed effects. In all wage specifications, the coefficients on unemployment exposure, non-participation exposure, and permanent income are statistically insignificant and often have signs opposite to the baseline results. We interpret this as evidence of limited within-family variation: siblings typically share nearly identical conditions during childhood. For instance, two siblings spaced two years apart share 16 of the same 18 years of maternal unemployment exposure. The implausible negative coefficient on permanent income, which contradicts both theory and other empirical evidence linking income positively to children's outcomes, reinforces the view that the fixed-effects specification lacks sufficient identifying variation.

We also assess the sensitivity of the unemployment coefficient to the permanent income coefficient. We run constrained regressions in which the coefficient on permanent income is fixed at different values within a pre-specified grid (Figure C5). For the association with unemployment exposure to disappear, the true coefficient on permanent income would need to be approximately five times larger than our OLS estimate and more than three times larger than our IV estimate. Such a value is implausibly large, indicating that income conditioning alone cannot account for the association between maternal unemployment exposure and children's adult wages.

Changes in Family Circumstances. Maternal unemployment during childhood may coincide with changes in family circumstances that are independently associated with children's outcomes. Table B4 adds controls for a range of family characteristics observed during childhood: a polynomial in the mother's age at childbirth, spousal labor supply, family structure and location (number of children, urban/rural residence), and the fraction of years in which fathers and grandparents were present. Extended family members have been shown to predict children's educational attainment and test scores (Loury, 2006). After accounting for these factors, the estimated association between maternal unemployment and adult wages is larger in magnitude and remains statistically significant.

We also account for local environments using restricted-use county geocode data to control for local labor market and community conditions. While county-level data are less granular than the neighborhood-level variation emphasized in Chetty et al. (2014, 2016, 2024), they provide meaningful information on local unemployment and income. Table B16 shows that county characteristics predict children’s future wages but do not account for the association between maternal unemployment exposure and adult outcomes.

Heterogeneity. Table B7 shows that maternal non-participation is associated with lower employment probability for both sons and daughters, with a larger association for daughters, consistent with the transmission of gender roles (Galassi, Koll, and Mayr, 2021). Maternal unemployment is associated with lower employment probability mainly for sons, with little association for daughters. No gender differences are found for wages. Table B8 shows that results do not differ substantially between children of single and married mothers. Figure C3 shows no heterogeneity by maternal education or AFQT score. Figure C4 shows that the negative associations persist as children age rather than dissipating over time.

Distinguishing Voluntary from Involuntary Non-Employment. Throughout most of the paper, we distinguish unemployment from non-participation. Unemployment is more likely to reflect involuntary separation, whereas non-participation can reflect voluntary choices (e.g., caregiving) or involuntary circumstances (e.g., discouraged workers). To separate these cases, we combine unemployment and non-participation into a single measure of non-employment and instrument it with children’s exposure to maternal employment in cyclical industries. Mothers employed in more cyclical industries are more likely to experience involuntary job loss due to industry downturns, so the predicted component of non-employment from this instrument captures the involuntary portion. The results in Table B10 show point estimates consistent with our main results but imprecise, likely due to weak instruments.

6 Conclusion

In this paper, we document the association between maternal unemployment during childhood and children’s adult labor-market outcomes, and unpack this relationship along several dimensions. Using linked mother-child data from the NLSY79 and NLSY79-CYA, we construct a continuous measure of cumulative exposure to maternal unemployment and relate it to children’s wages, employment, and educational attainment in adulthood.

Our first finding is that maternal unemployment exposure is a strong predictor of lower adult

wages and employment probability. We then decompose this association. The predictive power is concentrated in adolescence: exposure during ages 12-17 is associated with lower wages, earlier full-time labor-force entry, lower educational attainment, and sorting into lower-risk occupations, while early-childhood exposure has little predictive power for adult outcomes. Time-use and expenditure evidence from the ATUS and CE suggests that unemployed mothers increase time with children while reducing market-based inputs, consistent with input substitution partially offsetting the shock at young ages but not during adolescence, when educational investments and career decisions are central.

The association persists after conditioning on family permanent income, maternal ability proxies, and a rich set of family and local labor-market characteristics. Placebo tests using out-of-window maternal unemployment and bunching tests at zero exposure reduce concerns that the baseline patterns are driven by specific forms of selection on persistent unobserved maternal traits. Together, these results suggest that maternal unemployment exposure – particularly during adolescence – belongs among the early-life conditions that forecast adult labor-market outcomes. The persistence of the association after conditioning on observed family resources and included controls suggests that the relationship is not fully absorbed by measured income differences, leaving room for non-income mechanisms.

Our findings point to the adolescent years as the critical period through which the intergenerational costs of unemployment operate. Policies aimed at mitigating these costs may need to address not only contemporaneous income losses but also disruptions to children’s schooling and early labor-market transitions during this period.

References

- AGOSTINELLI, F. AND G. SORRENTI (2022): “Money vs. Time: Family Income, Maternal Labor Supply, and Child Development,” *Working Paper*.
- BAKER, M., J. GRUBER, AND K. MILLIGAN (2019): “The long-run impacts of a universal child care program,” *American Economic Journal: Economic Policy*, 11, 1–26.
- BASTIAN, J. AND L. LOCHNER (2022): “The EITC and maternal time use: more time working and less time with kids?” *Journal of Labor Economics*, 40, 573–611.
- BERNAL, R. (2008): “The effect of maternal employment and child care on children’s cognitive development,” *International economic review*, 49, 1173–1209.
- BILS, M. AND P. J. KLENOW (1998): “Using consumer theory to test competing business cycles models,” *Journal of Political Economy*, 106, 233–261.

- BILS, M., P. J. KLENOW, AND B. A. MALIN (2013): “Testing for Keynesian labor demand,” *NBER Macroeconomics Annual*, 27, 311–349.
- BINGLEY, P., L. CAPPELLARI, AND M. OVIDI (2024): “When it hurts the most: timing of parental job loss and a child’s education,” Unpublished manuscript.
- BLAU, F. D. AND A. J. GROSSBERG (1992): “Maternal Labor Supply and Children’s Cognitive Development,” *The Review of Economics and Statistics*, 74, 474–481.
- BONIN, H., T. DOHMEN, A. FALK, D. HUFFMAN, AND U. SUNDE (2007): “Cross-sectional earnings risk and occupational sorting: The role of risk attitudes,” *Labour Economics*, 14, 926–937.
- BRATBERG, E., Ø. A. NILSEN, AND K. VAAGE (2008): “Job losses and child outcomes,” *Labour Economics*, 15, 591–603.
- BRITTO, D. G. C., C. MELO, AND B. SAMPAIO (2023): “The Kids Aren’t Alright: Parental Job Loss and Children’s Outcomes Within and Beyond Schools,” Unpublished manuscript.
- CAETANO, C. (2015): “A test of exogeneity without instrumental variables in models with bunching,” *Econometrica*, 83, 1581–1600.
- CAETANO, C., G. CAETANO, H. FE, AND E. R. NIELSEN (2025): “A Dummy Test of Causal Identification in Models with Bunching,” Unpublished manuscript.
- CAETANO, C., G. CAETANO, E. NIELSEN, AND V. SANFELICE (2024): “The Effect of Maternal Labor Supply on Children: Evidence from Bunching,” Forthcoming at *Journal of Labor Economics*.
- CARD, D. (1999): “The causal effect of education on earnings,” *Handbook of Labor Economics*, 3, 1801–1863.
- CARNEIRO, P. AND J. HECKMAN (2003): “Human Capital Policy,” *IZA Discussion Papers*, IZA DP 821.
- CARNEIRO, P., K. V. LØKEN, AND K. G. SALVANES (2015): “A flying start? Maternity leave benefits and long-run outcomes of children,” *Journal of Political Economy*, 123, 365–412.
- CARNEIRO, P., K. G. SALVANES, B. WILLAGE, AND A. WILLÉN (2024): “Childhood shocks across ages and human capital formation,” Unpublished manuscript.
- CAUCUTT, E. M. AND L. LOCHNER (2020): “Early and late human capital investments, borrowing constraints, and the family,” *Journal of Political Economy*, 128, 1065–1147.
- CHETTY, R., W. S. DOBBIE, B. GOLDMAN, S. PORTER, AND C. YANG (2024): “Changing opportunity: Sociological mechanisms underlying growing class gaps and shrinking race gaps in economic mobility,” Unpublished manuscript.
- CHETTY, R., N. HENDREN, AND L. F. KATZ (2016): “The effects of exposure to better neighborhoods on children: New evidence from the moving to opportunity experiment,” *American Economic Review*, 106, 855–902.

- CHETTY, R., N. HENDREN, P. KLINE, AND E. SAEZ (2014): “Where is the land of opportunity? The geography of intergenerational mobility in the United States,” *The Quarterly Journal of Economics*, 129, 1553–1623.
- COELLI, M. B. (2011): “Parental job loss and the education enrollment of youth,” *Labour Economics*, 18, 25–35.
- COUCH, K. A. AND D. W. PLACZEK (2010): “Earnings losses of displaced workers revisited,” *American Economic Review*, 100, 572–589.
- DAHL, G. B. AND L. LOCHNER (2012): “The impact of family income on child achievement: Evidence from the earned income tax credit,” *American Economic Review*, 102, 1927–1956.
- DOUGAN, W. R., J. L. GARCÍA, AND I. POLOVNIKOV (2025): “Reexamining a Multisite Early Childhood Education Program: Gradual Scale-Up, Birth Spacing, and Parental Investment,” *Journal of Political Economy Microeconomics*. Accepted.
- DUSTMANN, C. AND U. SCHÖNBERG (2012): “Expansions in maternity leave coverage and children’s long-term outcomes,” *American Economic Journal: Applied Economics*, 4, 190–224.
- FRADKIN, A., F. PANIER, AND I. TOJEROW (2019): “Blame the parents? How parental unemployment affects labor supply and job quality for young adults,” *Journal of Labor Economics*, 37, 35–100.
- GALASSI, G., D. KOLL, AND L. MAYR (2021): “The Intergenerational Correlation of Employment: Mothers as Role Models,” *Working Paper*.
- HARTOG, J. AND W. P. VIJVERBERG (2007): “On compensation for risk aversion and skewness affection in wages,” *Labour Economics*, 14, 938–956.
- HEGARTY, C. (2025): “Parental Experience and Occupation Risk,” *Working Paper*.
- HILGER, N. G. (2016): “Parental job loss and children’s long-term outcomes: Evidence from 7 million fathers’ layoffs,” *American Economic Journal: Applied Economics*, 8, 247–283.
- JACOBSON, L. S., R. J. LALONDE, AND D. G. SULLIVAN (1993): “Earning Losses of Displaced Workers,” *The American Economic Review*, 83, 685–709.
- KRUEGER, A. B. AND A. I. MUELLER (2012): “Time use, emotional well-being, and unemployment: Evidence from longitudinal data,” *American Economic Review*, 102, 594–599.
- LINDO, J. M. (2011): “Parental job loss and infant health,” *Journal of health economics*, 30, 869–879.
- LOURY, L. D. (2006): “All in the extended family: Effects of grandparents, aunts, and uncles on educational attainment,” *American Economic Review*, 96, 275–278.
- MCLAUGHLIN, K. J. AND M. BILS (2001): “Interindustry mobility and the cyclical upgrading of labor,” *Journal of Labor Economics*, 19, 94–135.

- NECKER, S. AND A. VOSKORT (2014): “Intergenerational Transmission of Risk Attitudes—A revealed preference approach,” *European Economic Review*, 65, 66–89.
- OREOPOULOS, P., M. PAGE, AND A. H. STEVENS (2008): “The Intergenerational Effects of Worker Displacement,” *Journal of Labor Economics*, 26, 455–483.
- PEDTKE, J. H. (2026): “Family Structure and the Intergenerational Dynamics of Inequality,” Unpublished manuscript.
- RAPOSO, P., P. PORTUGAL, AND A. CARNEIRO (2021): “The sources of the wage losses of displaced workers: the role of the reallocation of workers into firms, matches, and job titles,” *Journal of Human Resources*, 56, 786–820.
- REGE, M., K. TELLE, AND M. VOTRUBA (2011): “Parental job loss and children’s school performance,” *The Review of Economic Studies*, 78, 1462–1489.
- STEVENS, A. H. (1997): “Persistent Effects of Job Displacement: The Importance of Multiple Job Losses,” *Journal of Labor Economics*, 15, 165–188.
- STEVENS, A. H. AND J. SCHALLER (2011): “Short-run effects of parental job loss on children’s academic achievement,” *Economics of Education Review*, 30, 289–299.
- UGUCCIONI, J. (2022): “The long-run effects of parental unemployment in childhood,” Unpublished manuscript.

A Data

A.1 NLSY Data Details

Our primary datasets are the National Longitudinal Survey of Youth 1979 (NLSY79) and the NLSY79 Child and Young Adult (NLSY79-CYA). The NLSY79 is a nationally representative sample of over 12,000 individuals aged 14-22 in 1979, providing a comprehensive longitudinal study of their lives and labor market experiences. The NLSY79-CYA contains information on children born to women in the original NLSY79 cohort, spanning from 1980 to 2018. The BLS has identified 11,551 children as having been born to the original 6,283 NLSY79 female respondents (as of 2018).

Table A1: Summary Statistics by Employment and Wage Availability

	Non-missing Emp Status	Non-missing Wage
Sample size	22,920	8,681
Unemployment Exposure	0.052 (0.074)	0.047 (0.070)
Non-participation Exposure	0.355 (0.295)	0.332 (0.281)
Permanent Income	0.798 (0.810)	0.857 (0.848)
Age	28.30 (4.63)	28.10 (4.27)
Log Wage	2.23 (0.44)	2.25 (0.42)
Log Hours	3.72 (0.20)	3.72 (0.19)

The Young Adult sample within the NLSY79-CYA is what enables us to study the long-term labor market impacts of maternal unemployment on children. The Young Adult sample refers to a questionnaire that has been administered to children aged 15 and older since 1994. The questionnaire was designed to facilitate life-cycle and cross-generational analyses. Of all identified children, 4,354 were interviewed as young adults. Details on the sizes and eligibility criteria vary by year. Those interested should look for details in the NLSY documentation.

Table A.1 has the descriptive statistics of our final samples. We include only young adults who have reached working age in our analysis. Column 1 shows the statistics for the respondents with non-missing values for employment status, child's gender, child's race, and child's age. Overall we have 22920 person-year non-missing employment observations. Column 2 shows the statistics for those who have non-missing values for wage, hours worked, total income, and child's gender, child's race, and child's age. There are 8682 person-year non-missing wage observations. The continuous variables display the mean and the standard deviation in parenthesis, while the categorical variables are described as fractions corresponding to each level.

UNEMP and OLF correspond to the exposures to maternal unemployment and non-participation in the labor force and are defined as in 2, taking values from 0 to 1. Permanent income is defined in 3 and is expressed in 10000s of 1993 US dollars. Age corresponds to the young adult’s age at the time of the survey. Log wage refers to a logarithm of hourly wages, while log hours – to a logarithm of weekly hours. Education and Mother’s Education indicate the highest attained education level for the young adults and their mothers respectively.

Notice, that Column 1 includes non-employed young adults, while Column 2 does not. This might explain why education levels are lower in Column 1. The rest of the variables have similar distributions across the two samples.

A.2 CES Data Details

The Current Employment Statistics (CES) data is a monthly survey conducted by the Bureau of Labor Statistics (BLS). It provides detailed information on employment, hours, and earnings of workers on nonfarm payrolls. Our exercise uses employment estimates for total nonfarm employment and by industry sector.

We use three-digit industries, but to describe the data in this appendix, we divide employment into 15 industries, as seen in Table A2. We use data spanning January 1972 to December 2019. Because, during these years, the US experienced significant changes in its sectorial composition, we show the total number of employees in each industry and their share of total employment in December 1972 and December 2019. The industries that gained employment share were financial activities, professional and business services, and private education and health services. Those that lost were construction, durable goods, and other services. We deliberately exclude the COVID-19 pandemic years from the sample.

We estimate the cyclical sensitivities of industries following the method of McLaughlin and Bils (2001). To account for the trend in sectorial composition, we use a cubic trend, which is flexible enough to capture it in these sectors. In the text, we mentioned that we estimate equation (5) in first-difference, but we actually use quarter changes, i.e., $E_{it} - E_{it-3}$. The reason is that we are interested in business cycle fluctuations, and we believe that monthly changes are too high-frequency. Ultimately, there is not much difference in the cyclical sensitivity β_i^e when using monthly or quarterly changes. Our estimated β_i^e ’s are displayed in Column 6.

A.3 ATUS Data Details

The American Time Use Survey (ATUS) is an annual survey conducted by the Bureau of Labor Statistics (BLS) to measure how the American population spends their time on different activities such as paid work, childcare, volunteering, and socializing. Participants are interviewed once and asked to describe their previous day’s activities, which are then categorized.

The ATUS sample is derived from households that have conducted their eighth interview for the Current

Table A2: Cyclical Sensitivities

	dez/1972		dez/2019		β_i^e
	Empl. (1,000)	Share	Empl. (1,000)	Share	
All industries	75,268	100.0%	151,666	100.0%	
Mining and logging	677	0.9%	708	0.5%	0.05
Construction	3,937	5.2%	7,541	5.0%	2.09
Durable goods	11,036	14.7%	8,022	5.3%	1.35
Nondurable goods	7,122	9.5%	4,794	3.2%	-0.06
Wholesale trade	3,578	4.8%	5,895	3.9%	0.14
Retail trade	8,227	10.9%	15,509	10.2%	-0.05
Transportation and warehousing	2,678	3.6%	5,736	3.8%	0.43
Utilities	564	0.7%	548	0.4%	-0.95
Information	2,096	2.8%	2,885	1.9%	0.24
Financial activities	3,842	5.1%	8,814	5.8%	-0.29
Professional and business services	5,668	7.5%	21,443	14.1%	0.21
Private education and health services	4,975	6.6%	24,391	16.1%	-0.77
Leisure and hospitality	5,240	7.0%	16,761	11.1%	-0.17
Other services	1,944	2.6%	5,911	3.9%	-0.49
Government	13,684	18.2%	22,709	15.0%	-0.85

Note: See text for details.

Population Survey (CPS). The ATUS survey selects roughly 25% of these households randomly to participate in the time-use survey. One individual aged 15 or above is chosen from each household to participate in the survey. Key demographic data collected in the CPS is transferred to the ATUS. This includes household membership, employment status, earnings, and other characteristics. We use data spanning 2003 and 2021 and focus on women between 25 and 50 who reported having children under 18 living in the household.

A.4 CE Data Details

Consumer Expenditure Surveys (CE) are conducted by the Census Bureau for BLS. For our analysis we use Interview Surveys of CE for 2000-2020, which collect information on monthly household expenditures, incomes, and household characteristics. Households are surveyed for five consecutive quarters and are then dropped from the sample. The survey collects detailed data on each household member, including income, employment status, education, gender, and relation to the respondent. We use this information to determine parental employment statuses and education levels.

The expenditures are recorded using Universal Classification Codes (UCC) and cover a wide range of personal consumption categories. We identify children-related expenditures based on whether the name of the expenditure contains one of the following children-related keywords: "infant", "nursery", "baby", "child", "boy", "girl", "school", "toys", "playground", "college", "tutoring". We then classify expenditures associated with babysitting, childcare, and daycare (UCC codes: 340210-340212, 670310) as childcare-related.

B Additional Tables

B.1 Robustness to Alternative Sample Selection Rules

In our baseline analysis, we define the main match as the job in which the respondent works the most hours at the survey interview date. Because the data are longitudinal, respondents may contribute multiple adult observations across survey waves. A related concern is that maternal unemployment affects educational attainment, which may in turn shift the age at which respondents first appear in the adult sample and the amount of labor-market experience they have when observed. This could mechanically affect observed wages and employment outcomes if more exposed individuals enter the sample earlier.

To assess the sensitivity of our results to how adult observations are constructed from the panel, we implement four alternative sample selection rules: (1) selecting only the first observed main match, so each respondent appears once; (2) selecting only the last observed main match, again with each respondent appearing once; (3) including all jobs across all surveys, allowing respondents to contribute multiple observations even within the same year; and (4) averaging job characteristics (such as wages and hours) across matches for each respondent-year.

Table B1 re-estimates equation (1) under these alternative selection rules. The estimated coefficients on maternal unemployment and labor-force non-participation remain similar in magnitude and statistical significance across specifications. These results suggest that our baseline findings are not driven by the particular way we construct adult job observations from the panel.

Table B1: Other Matches

	Main Match	First Match	Last Match	All Matches	Avg. of Matches
	(1)	(2)	(3)	(4)	(5)
	log(wage)	log(wage)	log(wage)	log(wage)	log(wage)
OLF	-0.121** (0.024)	-0.099** (0.024)	-0.095** (0.025)	-0.114** (0.023)	-0.097** (0.021)
UNEMP	-0.377** (0.076)	-0.446** (0.079)	-0.437** (0.085)	-0.345** (0.074)	-0.487** (0.076)
Permanent Income	0.105** (0.010)	0.094** (0.009)	0.111** (0.010)	0.099** (0.009)	0.108** (0.009)
Observations	8,376	4,116	4,116	10,943	4,269
R2	0.193	0.187	0.188	0.198	0.183

Note: This table presents robustness checks using alternative sample selection rules. Column (1) uses the baseline main match specification. Column (2) uses only the first observed match per respondent. Column (3) uses only the last observed match. Column (4) includes all jobs across all surveys. Column (5) averages job characteristics across all matches per respondent. UNEMP and OLF measure exposure to maternal unemployment and labor force non-participation, respectively (see equation 2). Permanent Income is measured in \$10,000s (discounted present value as of birth year; see equation 3). All specifications control for child demographics (race, gender, cubic in age) and survey year fixed effects. Estimates use ordinary least squares with standard errors clustered at the individual level (in parentheses).

* and ** indicate statistically significant at the 10% and 5% levels.

B.2 Non-Linear Effects on Wages and Employment Probabilities

Table B2 tests for non-linearity in the relationship between maternal non-employment and children's labor market outcomes. Rather than assuming linear effects, we divide exposure into quartiles and estimate equation (1) with dummy variables for each quartile (omitting the first quartile as the reference group). Because the distributions of unemployment and labor force non-participation exposure differ substantially, the quartile cutpoints vary by measure. For unemployment exposure, the 25th, 50th, and 75th percentiles correspond to approximately 0.04, 0.42, and 1.25 years. For non-participation exposure, these percentiles correspond to approximately 1.75, 5.25, and 9.83 years.

The results reveal different patterns for the two measures. For maternal unemployment (Columns 1-2), the OLS estimates show progressively larger negative effects across quartiles, with the top two quartiles (p50-75 and p75-100) negative and statistically significant in both columns; the p25-50 quartile is significant only in the OLS specification. For labor force non-participation (Columns 1-2), effects are more concentrated at the top of the distribution: the top quartile (p75-100) is significant in both OLS and IV, while the p50-75 quartile is significant only in the OLS specification.

The pattern differs slightly for employment probability (Columns 3-4). For unemployment exposure, the first two quartiles show no significant effect on children's employment probability, with negative effects emerging only in the top quartile. In Subsection 5.4, we report local polynomial regressions that further support approximate linearity for unemployment effects on wages.

Table B2: Exploring Non-linearity in the Effects of Maternal Labor-Force Non-Participation

	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
	log(wage)	log(wage)	emp.	emp.
OLF p25-50	-0.017 (0.017)	0.018 (0.018)	-0.015 (0.012)	-0.007 (0.012)
OLF p50-75	-0.042** (0.017)	0.002 (0.018)	-0.074** (0.012)	-0.064** (0.013)
OLF p75-100	-0.113** (0.018)	-0.046** (0.020)	-0.149** (0.012)	-0.109** (0.014)
UNEMP p25-50	-0.040** (0.017)	-0.007 (0.017)	-0.019 (0.012)	-0.016 (0.012)
UNEMP p50-75	-0.105** (0.018)	-0.061** (0.018)	-0.006 (0.013)	-0.005 (0.013)
UNEMP p75-100	-0.137** (0.019)	-0.072** (0.019)	-0.063** (0.013)	-0.050** (0.014)
Permanent Income		0.157** (0.022)		0.037** (0.013)
F statistic		58.428		105.206
Observations	9,203	9,203	22,920	21,737
R2	0.184	0.209	0.063	0.065

Note: This table presents non-linear specifications where unemployment (UNEMP) and labor force non-participation (OLF) exposure measures are replaced with quartile dummy variables. The omitted category is the first quartile (lowest exposure). Columns (1) and (3) show OLS estimates; Columns (2) and (4) show IV estimates where permanent income is instrumented using an alternative measure interacted with maternal characteristics. Dependent variables are log wages (Columns 1-2) and employment probability (Columns 3-4). Permanent Income is measured in \$10,000s (discounted present value as of birth year; see equation 3). All specifications control for child demographics (race, gender, education, cubic in age), survey year fixed effects, and maternal characteristics (marital status, primary earner status). Standard errors are clustered at the individual level (in parentheses).

* and ** indicate statistically significant at the 10% and 5% levels.

B.3 Home Environment and Child Development: Conditioning on Permanent Income

Table B3 reports IV estimates of the association between early-childhood maternal unemployment exposure and measures of home environment and cognitive development, conditioning on permanent income and correcting for measurement error as described in Subsection 3.6. These results complement the OLS estimates in Table 6.

Conditioning on permanent income substantially attenuates the associations with home environment quality. The UNEMP 0-5 coefficient on HOME Total decreases from -0.743 (OLS) to -0.256 (IV), and the Emotional subscale coefficient decreases from -0.859 to -0.327. The Cognitive subscale coefficient attenuates from -0.405 to -0.021. None of the IV estimates for unemployment exposure are statistically significant. The OLS coefficients are similarly attenuated and become insignificant after conditioning on permanent income.

The coefficient on permanent income itself is positive and statistically significant for HOME Total (0.220, standard error 0.056) and Cognitive stimulation (0.274, standard error 0.060), indicating that family resources are positively associated with home environment quality even after controlling for maternal education and cognitive ability. Permanent income also positively predicts average PIAT scores (0.177, standard error 0.067) but is not significantly associated with the Emotional subscale or Math PIAT.

As discussed in the main text, the IV estimates are imprecise. These results should therefore not be interpreted as establishing the absence of a relationship between early-childhood unemployment exposure and home environment quality. Rather, the informative comparison is the attenuation from OLS to IV: income loss accounts for a substantial portion of the unconditional association, suggesting that the income channel is the primary pathway through which early-childhood exposure relates to home environment measures.

Table B3: Association with Home Environment and Child Development: Conditioning on Permanent Income

	(1)	(2)	(3)	(4)	(5)
	HOME Total	Emotional	Cognitive	Avg. PIAT	Math PIAT
OLF 0-5	-0.054 (0.062)	-0.042 (0.071)	-0.028 (0.068)	0.000 (0.069)	-0.036 (0.067)
UNEMP 0-5	-0.256 (0.211)	-0.327 (0.217)	-0.021 (0.236)	0.035 (0.209)	0.180 (0.217)
Permanent Income 0-5	0.220** (0.056)	0.079 (0.060)	0.274** (0.060)	0.177** (0.067)	0.091 (0.063)
Mom Educ Dummies	Yes	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes	Yes
F statistic	36.724	36.914	35.740	40.509	40.509
Observations	5,846	5,577	5,713	5,641	5,641
R2	0.340	0.262	0.253	0.172	0.169

Note: OLF and UNEMP are our exposure measures to maternal labor-force non-participation and unemployment, respectively, measured during ages 0-5. See equation (2) for construction details. Permanent Income is in \$10,000s (discounted present value) as of birth year, measured during ages 0-5. See equation (3) for construction details. Controls include maternal education attainment and AFQT quintile dummies, dummy variables for children's race and gender, a cubic polynomial in children's age, and survey-year fixed effects. Dependent variables are total HOME score, emotional support HOME score, cognitive stimulation HOME score, average PIAT score, and PIAT Math score, all measured between ages 7 and 9. The average PIAT score is an average of the PIAT Math, PIAT Reading Recognition, and Reading Comprehension scores. All models are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses. The F statistic is the Kleibergen-Paap F statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

B.4 Additional Controls

In our baseline analysis, we control for permanent income, child demographics, maternal characteristics, and three proxies for maternal ability (education, AFQT score, and unemployment outside ages 0-18). Table B4 assesses robustness to including additional controls capturing other aspects of children’s environment during childhood.

Column 1 replicates our baseline IV specification from Table 8, Column 4. Column 2 adds a polynomial in the mother’s age at childbirth. Column 3 controls for spousal labor supply (average weekly hours and weeks worked). Column 4 includes family structure and location variables (average number of children in the household, urban/rural residence). Column 5 adds the fraction of years that grandparents and fathers were present in the household. Column 6 includes all additional controls simultaneously.

The results show that our findings are robust—indeed, strengthened—by these additional controls. The estimated effect of maternal unemployment becomes more negative, increasing in magnitude from -0.21 (SE 0.08) to -0.32 (SE 0.09) when all controls are included, and remains highly significant. This pattern suggests that the additional controls capture positive correlates of both maternal unemployment and child wages (such as extended family support), so controlling for them reveals a larger negative causal effect. By contrast, the coefficient on maternal labor force non-participation remains small and statistically insignificant across all specifications, consistent with our interpretation that this measure captures heterogeneous behaviors that do not have consistent causal effects on children.

Table B4: Robustness: Additional Controls

	Baseline	w/ Mom's Age	Mom's Spouse	Family Envir.	HH Composition	All Controls
	(1)	(2)	(3)	(4)	(5)	(6)
	log(wage)	log(wage)	log(wage)	log(wage)	log(wage)	log(wage)
OLF	-0.035 (0.026)	-0.033 (0.026)	-0.026 (0.028)	-0.037 (0.026)	-0.034 (0.026)	-0.025 (0.028)
UNEMP	-0.214** (0.078)	-0.214** (0.078)	-0.324** (0.088)	-0.214** (0.078)	-0.212** (0.078)	-0.323** (0.087)
Permanent Income	0.102** (0.024)	0.101** (0.025)	0.108** (0.025)	0.102** (0.026)	0.103** (0.024)	0.104** (0.027)
Educ Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes	Yes	Yes
Add. unemp	Yes	Yes	Yes	Yes	Yes	Yes
Mom Cubic in Age		Yes				Yes
Mom' Spouse Labor Supply			Yes			Yes
Family Environment				Yes		Yes
Household Composition					Yes	Yes
F statistic	49.962	49.199	48.614	44.030	49.383	40.467
Observations	7,707	7,707	6,905	7,695	7,707	6,893
R2	0.273	0.274	0.268	0.276	0.273	0.272

Note: This table presents robustness checks with additional control variables. The dependent variable is log wages. Column (1) replicates the baseline IV specification from Table 8, Column 4. Additional controls are: (2) cubic polynomial in mother's age at childbirth; (3) spouse's average weekly hours and weeks worked; (4) average number of children in household and urban/rural residence; (5) fraction of years with grandparents and father present; (6) all additional controls simultaneously. All specifications include baseline controls: permanent income, child demographics, maternal characteristics, and maternal ability proxies (education, AFQT, unemployment outside ages 0-18). All models estimated by two-stage least squares, instrumenting permanent income with an alternative measure interacted with maternal characteristics. Standard errors clustered at the individual level (in parentheses). The F statistic is the Kleibergen-Paap F statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

B.5 Number of Unemployment Spells

Our baseline analysis measures total exposure to maternal unemployment during childhood (ages 0-18). We now decompose this total exposure into two components: the number of unemployment spells and the average duration per spell. We define an unemployment spell as a transition from any labor market status (employment or out-of-labor-force) to unemployment. For each mother, we count the total number of such transitions during the child's first 18 years. Average duration per spell is calculated as total unemployment exposure divided by the number of spells (defined as zero for mothers with no unemployment spells). Note that by construction, total exposure equals the number of spells times average duration per spell.

Table B5 presents the results. Column 1 shows that a greater number of maternal unemployment spells is significantly associated with lower children's wages: each additional spell reduces wages by 0.7% (standard error 0.003). The number of out-of-labor-force spells shows no significant relationship. Column 2 estimates the effect of average duration per spell while controlling for permanent income and other baseline controls. The coefficient on average unemployment duration is negative (-0.359) but imprecisely estimated (standard error 0.218). Lastly, in Column 3, we include both measures simultaneously. The coefficient on the number of unemployment spells remains negative and statistically significant, while the coefficient on average unemployment exposure remains negative but imprecisely estimated. This suggests that frequency rather than duration per spell may be the more important mechanism.

Table B5: Robustness: Number of Unemployment Spells

	(1)	(2)	(3)
	log(wage)	log(wage)	log(wage)
Number of OLF Spells	0.001 (0.002)		0.000 (0.002)
Average OLF		-0.020 (0.050)	-0.061 (0.053)
Number of UNEMP Spells	-0.007** (0.003)		-0.007** (0.003)
Average UNEMP		-0.359 (0.305)	-0.254 (0.301)
Permanent Income	0.112** (0.024)	0.117** (0.023)	0.103** (0.024)
Educ Dummies	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes
Remaining Unemp	Yes	Yes	Yes
F statistic	66.670	77.717	61.233
Observations	7,672	7,672	7,672
R2	0.273	0.270	0.274

Note: See Tables 8 for the definition of mothers' ability control and for the construction of other variables. See text for additional controls. All models are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses. The F statistic is the Kleibergen-Paap F statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

B.6 Fixed-Effect Results

Our baseline analysis exploits cross-sectional variation across children. The NLSY's panel structure also allows comparisons of siblings who were exposed to different levels of maternal unemployment during childhood. To do so, we estimate equation (1) with mother fixed effects, so identification comes only from within-family differences in exposure. Table B6 reports these estimates. Columns 1-2 present OLS results, and Columns 3-4 present IV results.

The fixed-effect estimates do not corroborate the baseline cross-sectional patterns. Across specifications, the coefficients on unemployment exposure, labor-force non-participation, and permanent income are statistically insignificant, and several estimates are imprecisely estimated with unstable signs. We therefore view this exercise as a cautionary robustness check rather than as supporting evidence for the baseline estimates.

There are also reasons to expect the mother fixed-effect design to be weak in this setting. Siblings share much of their childhood environment and often have similar exposure histories. For example, siblings born two years apart overlap for 16 of the 18 childhood years used to construct our exposure measures. This overlap limits the within-family variation on average. The limited within-family variation likely contributes to the low precision of the fixed-effect estimates.

We interpret the fixed-effect results as showing that within-family comparisons do not provide a useful corroboration of the baseline relationship in our data. The estimates are imprecise and unstable, and the identifying variation is limited. For this reason, our main analysis relies on the cross-sectional specifications with rich controls rather than on sibling comparisons.

Table B6: Fixed-Effect Results: Wages

	OLS		IV	
	(1) log(wage)	(2) log(wage)	(3) log(wage)	(4) Emp.
OLF	0.055 (0.094)	0.045 (0.094)	0.023 (0.072)	-0.035 (0.050)
UNEMP	0.260 (0.285)	0.297 (0.293)	0.340 (0.238)	-0.099 (0.140)
Permanent Income	-0.029 (0.046)	-0.016 (0.047)	0.160 (0.136)	0.185** (0.077)
Educ Dummies		Yes	Yes	Yes
Mom Controls		Yes	Yes	Yes
F statistic			54.327	229.878
Observations	8,680	8,287	7,350	20,565
R2	0.678	0.691	0.156	0.022

Note: Column 1 includes dummy variables for children's race, a cubic polynomial in children's age, and fixed effects for the survey year. Column 2 includes, in addition, dummy variables for educational attainment, mothers' marital status, and primary earner status. These equations were estimated by ordinary least squares. Columns 3 and 4 include dummy variables for children's race, a cubic polynomial in children's age, fixed effects for the survey year, children's educational attainment dummies, and mothers' marital and primary earner status dummies. These equations were estimated by two-stage least squares. Excluded instruments used in all columns are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. All columns include mother-fixed effects. Standard errors are reported in parentheses. They are clustered at the children's level for Columns 1 to 2 but not for Columns 3 and 4. Clustering made the estimated covariance matrix not of full rank. The F statistic is the Cragg-Donald Wald F statistic, which assumes that the errors are independent and identically distributed. This might not be true in our context, and the results should be interpreted with caution.

* and ** indicate statistically significant at the 10% and 5% levels.

B.7 Effects by the Gender of the Child

Our baseline analysis assumes homogeneous effects across children of different genders. However, if maternal employment patterns influence children through role modeling or gender-specific socialization, effects may differ by child gender. To test for such heterogeneity, we re-estimate equation 1 including interactions between child gender and maternal unemployment and labor force non-participation exposure.

Table B7 presents the results. Columns 1 and 3 replicate the baseline estimates without interactions, while Columns 2 and 4 include interactions with child gender. Column 2 shows no significant gender differences in the effects of maternal non-employment on children's wages: both sons and daughters experience similar wage penalties from maternal unemployment exposure.

Column 4 suggests gender-specific patterns for employment probabilities. The main OLF coefficient (-0.057, SE 0.024) captures the effect for sons and is negative and statistically significant. For daughters, the relevant effect is the sum of the main term and the interaction, equal to $-0.057 - 0.152 = -0.209$, implying a larger negative association with employment. Similarly, the main UNEMP coefficient (-0.293, SE 0.096) captures the effect for sons. For daughters, the implied effect is $-0.293 + 0.291 = -0.002$, which is close to zero. Taken together, these estimates suggest that maternal out-of-the-labor-force exposure is associated with lower employment probabilities for both sons and daughters, whereas maternal unemployment appears to reduce employment mainly for sons. These patterns may reflect gender-norm transmission, with daughters of non-working mothers being more likely to remain out of work, while sons may be more affected by the income and stress channels associated with maternal unemployment. However, we view these results as exploratory.

Table B7: Differences by Gender

	(1)	(2)	(3)	(4)
	log(wage)	log(wage)	emp.	emp.
OLF	-0.035 (0.026)	-0.043 (0.034)	-0.132** (0.019)	-0.057** (0.024)
UNEMP	-0.214** (0.078)	-0.210* (0.111)	-0.137* (0.070)	-0.293** (0.096)
Permanent Income	0.102** (0.024)	0.103** (0.024)	0.008 (0.014)	0.008 (0.014)
Female	-0.103** (0.012)	-0.108** (0.019)	-0.054** (0.009)	-0.018 (0.014)
Female # OLF		0.017 (0.043)		-0.152** (0.032)
Female # UNEMP		-0.006 (0.136)		0.291** (0.121)
Educ Dummies	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes
Add. unemp	Yes	Yes	Yes	Yes
F statistic	49.962	50.161	77.825	78.168
Observations	7,707	7,707	20,102	20,102
R2	0.273	0.273	0.094	0.097

Note: See Tables 8 for the definition of mothers' ability control and for the construction of other variables. Controls include dummy variables for children's race, and educational attainment, a cubic polynomial in children's age, and fixed effects for the survey year, mothers' marital and primary earner statuses. All columns are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

B.8 Effects by the Single Mom

We use household composition variables to measure the presence of a spouse in the household over time. Specifically, we count the number of survey waves in which the respondent's spouse was present and divide the sample into two groups: (i) respondents with spousal presence frequency below the median and (ii) respondents with spousal presence frequency above the median. We then estimate separate regressions for each group. Table B8 shows no substantial difference in the unemployment coefficient across groups, suggesting that having a spouse present does not significantly alter the long-term effects of maternal unemployment on wages.

Table B8: Differences by Single Mom

	No Spouse Present	Spouse Present
	(1)	(2)
	log(wage)	log(wage)
OLF	-0.159** (0.031)	-0.111** (0.016)
UNEMP	-0.142* (0.084)	-0.327** (0.076)
Permanent Income	0.030 (0.027)	0.003 (0.011)
Educ Dummies	Yes	Yes
F statistic	163.124	.
Observations	11,019	9,529
R2	0.087	0.076

Note: See Tables 8 for the definition of mothers' ability control and for the construction of other variables. Controls include dummy variables for children's race and educational attainment, a cubic polynomial in children's age, fixed effects for the survey year, and dummies for primary earner status. We do not include marital status dummies. All columns are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

B.9 Birth and Prenatal Characteristics

We examine whether birth and prenatal characteristics – birth order, gestation length, birth weight, and birth spacing among siblings – affect our estimates of the impact of maternal unemployment on children’s future wages. Including birth spacing is motivated by evidence from [Dougan et al. \(2025\)](#), who show that the effectiveness of the Infant Health and Development Program (IHDP) depends on whether a child is a twin and, more broadly, on the spacing between births. As shown in Table B9, controlling for these factors does not alter the estimated negative and statistically significant effect of maternal unemployment. Most birth-related controls are small and insignificant, except for existence of sibling, which shows a modest positive effect. These results indicate that early-life conditions do not confound our main finding.

Table B9: Birth and Prenatal Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)
	log(wage)	log(wage)	log(wage)	log(wage)	log(wage)	log(wage)
OLF	-0.035 (0.026)	-0.036 (0.026)	-0.040 (0.027)	-0.044* (0.027)	-0.037 (0.026)	-0.044 (0.027)
UNEMP	-0.213** (0.078)	-0.214** (0.078)	-0.221** (0.081)	-0.231** (0.079)	-0.216** (0.078)	-0.227** (0.082)
Permanent Income	0.103** (0.024)	0.108** (0.026)	0.099** (0.025)	0.100** (0.025)	0.111** (0.026)	0.109** (0.027)
Birth order of child		0.010 (0.007)				0.005 (0.008)
Length of gestation (weeks)			0.004 (0.003)			0.003 (0.003)
Birth weight (ounces)				0.000 (0.000)		0.000 (0.000)
Has siblings					0.058** (0.028)	0.050* (0.029)
Has siblings=1 × Birth space					0.002 (0.002)	0.001 (0.002)
Educ Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes	Yes	Yes
Remaining Unemp	Yes	Yes	Yes	Yes	Yes	Yes
F statistic	50.051	44.996	46.226	46.400	41.934	35.840
Observations	7,713	7,713	7,214	7,409	7,713	7,068
R2	0.273	0.272	0.277	0.271	0.273	0.275

Note: See Tables 8 for the definition of mothers’ ability control and for the construction of other variables. All columns are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers’ marital and primary earner statuses. Standard errors are clustered at the children’s level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

B.10 Distinguishing Between Voluntary and Involuntary Unemployment

Throughout most of the paper, we have examined two labor market statuses: unemployment and non-participation in the labor force. Unemployment is usually involuntary and caused by external factors beyond an individual's control; however, non-participation can be either voluntary or involuntary. For example, non-participation is voluntary if mothers choose not to work for personal reasons, such as caring for children or other family members. It can be involuntary if mothers are not employed and have stopped looking for work due to reasons beyond their control, such as inability to find suitable employment, lack of job opportunities, or disability.

Some of the mothers in our sample might have self-selected to not participate in the labor market, raising concerns about the interpretation of the coefficient on the non-participation measure. In this subsection, we identify the effect of exposure to involuntary labor-market non-participation on children's outcomes. We combine our unemployment and non-participation exposure measures into a single measure of non-employment exposure and instrument it with children's exposure to maternal employment in cyclical industries.

The idea is that mothers who work in more cyclical industries are more likely to experience involuntary job loss due to factors beyond their control. We interpret a mother's non-employment predicted by her industry's cyclical exposure as the involuntary portion of her non-employment. To construct instruments for this exercise, we follow these steps: (1) we calculate a measure of cyclical sensitivity β_i^e for each industry i , (2) using the NLSY labor history array, we associate each job with its industry's measure of cyclical sensitivity, (3) we multiply the cyclical sensitivity measure by a business cycle indicator for each month, and (4) we take the average cyclical sensitivity that each child was exposed to during childhood.

We use two measures of cyclical sensitivities of industries. First, we follow the method of [McLaughlin and Bils \(2001\)](#) and project the industry i employment E_{it} on aggregate employment E_t and a cubic trend,

$$\ln(E_{it}/E_t) = a_i + a_{1i}t + a_{2i}t^2 + a_{3i}t^3 + \beta_i^e \ln E_t + e_{it} . \quad (5)$$

Our measure of industry i 's cyclical sensitivity is β_i^e . We estimate equation (5) for each industry i in the first difference. The employment variables come from the Current Employment Statistics (CES) data, a survey conducted monthly by the Bureau of Labor Statistics (BLS). We use three-digit industries in the estimation. Our time series spans from 1972 until 2019. We give a more detailed data description in the [Appendix A](#).

Our second measure of industries' cyclical sensitivities is the durability of the goods associated with them. According to consumer theory, the durability of goods should predict the cyclical behavior of expenditures on these goods. This is because durable goods require substantial investment to increase their stock, with small increases in stock potentially leading to a proportionally larger increase in spending. Consequently, the demand for durable goods is highly sensitive to the economic cycle. This prediction is confirmed by [Bils and Klenow \(1998\)](#) and [Bils, Klenow, and Malin \(2013\)](#). We utilize durability measures for seventy goods, as constructed by [Bils et al. \(2013\)](#). These measures are derived from data sourced from the US Na-

tional Income and Product Accounts (NIPA) and estimates provided by an insurance company.

We use two instruments for each cyclical measure. The first instrument is the job's cyclical sensitivity, where simply working in a more cyclical job is assumed to predict higher involuntary non-employment, regardless of business cycle conditions. The second instrument is the job's cyclical sensitivity interacted with a business cycle measure. For our baseline estimate, we use annual real GDP growth as our measure of the business cycle, where working in a cyclical job during periods of negative growth is assumed to predict involuntary non-employment.

In Table B10, Column 2 reports results using the job's cyclical sensitivity and its interaction with the GDP growth for predicting involuntary non-employment, while Column 3 uses the durability measure and its interaction with the GDP growth. Column 4 uses both the job's cyclical sensitivity and the durability measure. For comparison, Column 1 shows the results without instrumenting for involuntary non-employment. The coefficients from the instrumented regressions are significantly larger than those from the non-instrumented regressions. Moreover, the coefficients from the instrumented regressions are statistically significant when the job's cyclical sensitivity is used as an instrument in Column 2. This is our preferred specification. The specification that uses only the durability measure is not significant, potentially given large standard errors driven by a weak-IV problem. However, the point estimate is in line with the other columns.

The coefficient on non-employment exposure in Table B10, Column 2 is -0.171, similar to the coefficient of -0.207 on unemployment exposure in Table 8, Column 5. This stark similarity, when accounting for involuntary non-participation, implies that maternal involuntary non-employment also has scarring effects on children's future labor market outcomes. Specifically, under the assumption that all non-employment is involuntary, these results suggest that a child exposed to one standard deviation higher involuntary maternal non-employment is estimated to experience a 7.0% reduction in adult wages. Lastly, Table B10, Column 5 reports the employment specification.

Table B10: Using Exposure to Cyclical Industries to Disentangle Voluntary and Involuntary OLF: Effects on Wages

	(1)	(2)	(3)	(4)	(5)
	log(wage)	log(wage)	log(wage)	log(wage)	emp.
OLF + UNEMP	-0.049*	-0.171*	-0.060	-0.110	-0.182**
	(0.026)	(0.101)	(0.099)	(0.092)	(0.083)
Permanent Income	0.103**	0.055	0.101**	0.081**	-0.010
	(0.025)	(0.045)	(0.044)	(0.041)	(0.033)
Inst. for Measurement Error	Yes	Yes	Yes	Yes	Yes
Cyclical-Job Measure		Yes		Yes	Yes
Durability Measure			Yes	Yes	Yes
F statistic	50.357	9.791	10.252	8.806	7.222
Observations	7,527	7,527	7,527	7,527	19,486
R2	0.270	0.270	0.271	0.272	0.089

Note: OLF + UNEMP are our exposure measures to maternal non-employment. See equation (2) for construction details. Permanent Income is in \$10,000s (discounted present value) as of birth year. See equation (3) for construction details. Controls include dummy variables for children's race, gender, and educational attainment, a cubic polynomial in children's age, dummy variables for mothers' marital and primary earner statuses, mothers' ability controls, and fixed effects for the survey year. See Tables 8 and 8 for the definition of mothers' ability control. All models are estimated by two-stage least squares. Excluded instruments used in all Columns are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Columns 2, 3, and 4 use different measures of exposure to cyclical industries as additional instruments. See text for details. Standard errors are clustered at the children's level and reported in parentheses. The F statistic is the Kleibergen-Paap F statistic.

⁺, ^{*}, and ^{**} indicate statistically significant at the 15%, 10%, and 5% levels.

B.11 Other Labor Market Outcomes: Occupation Return and Risk

We documented that exposure to maternal unemployment and labor-market non-participation predicts children's lower earnings. It is plausible to imagine that a child who is exposed to maternal unemployment self-selects into safer occupations, presumably because the child might develop a risk aversion against earning risk. [Hegarty \(2025\)](#) finds evidence of the exact mechanism using the Panel Study of Income Dynamics (PSID). She constructs a measure of lifetime earnings risk for 22 occupations and documents that parental layoffs are correlated with children earning less in their early careers and working in occupations with lower risk.

In this subsection, we document that children with more exposure to maternal unemployment work in low-risk occupations, but we do not find that these occupations have low returns. We measure occupational earning return and risk using Mincer equation regressions following [Bonin et al. \(2007\)](#), [Hartog and Vijverberg \(2007\)](#), and [Necker and Voskort \(2014\)](#). First, we stack the March Current Population Survey (CPS) data from 1998 to 2021. We use the Autor-Dorn crosswalk to harmonize occupation codes across years. Second, we regress the logarithm of wages on a cubic polynomial in potential experience, educational attainment dummies, sex and race dummies, year-fixed effects, and occupation-fixed effects. Our measure of occupation mean returns is the estimated coefficients on the occupation dummies. Our measure of occupation risk is the standard deviation of the residuals within each occupation. Our sample includes occupations with at least 100 individuals and normalizes the measures by their standard deviation. Therefore, the regression coefficient should be interpreted as the impact of working in an occupation with one standard deviation above the average return or risk.

Table [B11](#) shows the results when we use 3-digit occupation codes to create our measures. Columns 1 and 2 show that occupation returns are unrelated to the unemployment exposure measure. Columns 3 and 4 show that more unemployment exposure is associated with lower occupational earning risk. As mentioned, this finding is consistent with [Hegarty \(2025\)](#) and the mechanism that exposure to unemployment or parental layoffs during childhood leads individuals to self-select into lower-earning but safer occupations. In Column 4, we focus on individuals over 30 years old. Since we interpret the previous result as related to risk aversion, we expect a stronger effect for older workers (i.e., even lower occupational risk). Job search and career transitions take time, so older workers had more opportunities to self-select into occupations consistent with their risk preferences. Consistent with this interpretation, we find a stronger effect for this subgroup.

Our baseline measure of occupational earning risk is constructed using 3-digit occupations. Table [B12](#) uses 2-digit occupations codes. In Columns 1 and 2, we continue to find occupation returns that are unrelated to the unemployment exposure measure. In Columns 3 and 4, we observe that the impact of unemployment exposure is still negative but not significant. Moreover, the result is stronger for the older sample but, again, not significant. We conclude that having finer occupation codes is essential for the documented result.

Table B11: Other Labor Market Outcomes: Earning Risk

	All Sample	Above 30 Yr	All Sample	Above 30 Yr
	(1)	(2)	(3)	(4)
	3d-Occ Return	3d-Occ Return	3d-Occ Risk	3d-Occ Risk
OLF	-0.007 (0.015)	-0.035 (0.025)	0.049 (0.070)	0.028 (0.111)
UNEMP	0.044 (0.051)	0.057 (0.085)	-0.520** (0.210)	-0.960** (0.269)
Permanent Income	0.028** (0.013)	0.014 (0.024)	0.009 (0.060)	-0.028 (0.098)
Educ Dummies	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes
Add. unemp	Yes	Yes	Yes	Yes
F statistic	49.403	18.169	48.430	18.066
Observations	7,517	2,191	7,501	2,194
R2	0.224	0.183	0.030	0.047

Note: See text for details how how earning risk is constructed. See Tables 8 for the definition of mothers' ability control and for the construction of other variables. All columns are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

Table B12: Other Labor Market Outcomes: Earning Risk

	All Sample	Above 30 Yr	All Sample	Above 30 Yr
	(1)	(2)	(3)	(4)
	2d-Occ Return	2d-Occ Return	2d-Occ Risk	2d-Occ Risk
OLF	-0.004 (0.013)	-0.035 (0.024)	0.070 (0.066)	0.034 (0.109)
UNEMP	0.010 (0.050)	0.037 (0.083)	-0.322 (0.228)	-0.480 (0.310)
Permanent Income	0.019 (0.012)	0.015 (0.024)	0.084 (0.060)	0.082 (0.091)
Educ Dummies	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes
Add. unemp	Yes	Yes	Yes	Yes
F statistic	50.051	18.731	50.048	18.731
Observations	7,713	2,248	7,711	2,248
R2	0.207	0.163	0.025	0.054

Note: See text for details how how earning risk is constructed. See Tables 8 for the definition of mothers' ability control and for the construction of other variables. All columns are estimated by two-stage least squares. Excluded instruments are the alternative permanent income measure and its interactions with dummy variables for mothers' marital and primary earner statuses. Standard errors are clustered at the children's level and reported in parentheses.

* and ** indicate statistically significant at the 10% and 5% levels.

B.12 Additional Results on Mothers' Time of Use

Table B13: Average Number of Minutes per Mother' Status and Family Income

	Emp	Unemp	OLF
Total Time Spent on All Activities	1440.00	1440.00	1440.00
Caring for and Helping Household Members	109.38	162.20	205.07
Caring for and Helping Children	88.16	138.46	179.15
Education-Related Activities for Children	10.28	17.50	23.12
Health-Related Activities for Children	3.07	5.80	4.98
Other Caring Activities for Children	74.80	115.15	151.05
Caring for and Helping Non-Household Members	6.55	14.21	11.11
Working and Work-Related Activities	385.55	38.41	4.90
Leisure and Social Activities	173.24	271.58	258.46
Purchasing Goods and Eating	102.14	135.82	132.13
Personal Care and Household Activities	635.74	761.39	774.72
Educational Activities	7.43	22.38	19.28
Other and Communication Activities	19.96	34.02	34.33

Note: The sample consists of women between 25 and 50 who reported having children under 18 living in the household. We look at the days of the week and non-holidays. Time allocation categories are the major BLS aggregate categories. Each cell represents averages. We use survey sample weights and construct additional weights to correct for differences in family income across employment status groups.

Table B14: Average Number of Minutes per Mother' Status and School Groups

	Emp	Unemp	OLF
Total Time Spent on All Activities	1440.00	1440.00	1440.00
Caring for and Helping Household Members	106.88	161.73	204.20
Caring for and Helping Children	85.93	138.58	178.52
Education-Related Activities for Children	10.24	18.08	22.77
Health-Related Activities for Children	2.97	5.75	5.17
Other Caring Activities for Children	72.72	114.74	150.58
Caring for and Helping Non-Household Members	6.76	14.17	11.37
Working and Work-Related Activities	384.04	37.10	4.93
Leisure and Social Activities	174.26	271.48	259.80
Purchasing Goods and Eating	101.81	132.47	131.46
Personal Care and Household Activities	639.88	761.37	774.19
Educational Activities	6.78	27.76	19.99
Other and Communication Activities	19.59	33.91	34.06

Note: The sample consists of women between 25 and 50 who reported having children under 18 living in the household. We look at the days of the week and non-holidays. Time allocation categories are the major BLS aggregate categories. Each cell represents averages. We use survey sample weights and construct additional weights to correct for differences in education levels across employment status groups.

Table B15: Average Number of Minutes per Mother' Status, Age of the Youngest Child, and the Number of Children

	Emp	Unemp	OLF
Total Time Spent on All Activities	1440.00	1440.00	1440.00
Caring for and Helping Household Members	113.85	153.17	183.77
Caring for and Helping Children	92.25	130.47	158.66
Education-Related Activities for Children	10.33	18.68	22.74
Health-Related Activities for Children	3.22	4.63	4.70
Other Caring Activities for Children	78.70	107.16	131.21
Caring for and Helping Non-Household Members	6.46	14.72	12.62
Working and Work-Related Activities	383.45	36.86	5.33
Leisure and Social Activities	171.74	277.99	271.00
Purchasing Goods and Eating	103.09	125.94	131.12
Personal Care and Household Activities	634.10	771.80	781.20
Educational Activities	7.24	28.13	21.12
Other and Communication Activities	20.08	31.40	33.84

Note: The sample consists of women between 25 and 50 who reported having children under 18 living in the household. We look at the days of the week and non-holidays. Time allocation categories are the major BLS aggregate categories. Each cell represents averages. We use survey sample weights and construct additional weights to correct for differences in the age of the youngest child and the number of children across employment status groups.

B.13 Local Labor Market and Mobility

Chetty et al. (2014), Chetty et al. (2016), and Chetty et al. (2024) document the importance of neighborhood environments in shaping children’s long-term outcomes. To account for this channel, we use restricted-use County Geocode data to ensure that our findings are not simply driven by geographic location. While our data is limited to the county level – less granular than the neighborhood-level variation emphasized in prior work – it still allows a meaningful examination of the local environment. These geocode files provide detailed locational information on survey respondents, including their county of residence at various points in time. This allows us to complement our analysis of maternal unemployment exposure with an investigation into how county-level conditions influence children’s future labor market outcomes.

We construct three measures of the local environment during childhood: (1) Exposure to County Unemployment, defined as the average county-level unemployment rate during the child’s formative years; (2) Exposure to County Income, measured using per capita personal income; and (3) Number of Counties Lived In, capturing the extent of residential mobility, which may reflect instability and have independent effects on child development. County unemployment data come from the Bureau of Labor Statistics’ Local Area Unemployment Statistics (LAUS), which provide monthly estimates of labor force conditions at the county level. County income data are obtained from the Bureau of Economic Analysis’ Regional Economic Accounts, which report annual measures of personal income and population.

Table B.13 shows that county-level unemployment and income are both predictive of children’s future wages. However, they are insufficient to account for the negative impact of maternal unemployment exposure, suggesting that the mechanisms at play go beyond local labor market conditions. Columns 1 and 3 show the baseline results for comparison.

Table B16: Robustness: Controlling for Local Labor Market and Mobility

	(1)	(2)	(3)	(4)
	log(wage)	log(wage)	log(wage)	log(wage)
OLF	-0.037 (0.026)	-0.056** (0.026)	-0.035 (0.026)	-0.033 (0.026)
UNEMP	-0.216** (0.078)	-0.198** (0.077)	-0.213** (0.078)	-0.211** (0.078)
Permanent Income	0.099** (0.024)	0.067** (0.026)	0.103** (0.024)	0.103** (0.024)
County Unemp		0.011** (0.003)		
County Income		0.281** (0.037)		
Number of moves				-0.004 (0.006)
Educ Dummies	Yes	Yes	Yes	Yes
Mom Educ Dummies	Yes	Yes	Yes	Yes
Mom AFQT	Yes	Yes	Yes	Yes
Remaining Unemp	Yes	Yes	Yes	Yes
F statistic	49.423	44.010	50.051	50.100
Observations	7,652	7,652	7,713	7,713
R ²	0.273	0.289	0.273	0.273

Notes: OLF and UNEMP capture exposure to maternal non-participation and unemployment, respectively (see equation 2). Permanent Income is measured in \$10,000s as the discounted value at birth year (see equation 3). County-level unemployment and income are included to account for local labor market conditions. All models control for children's demographics and education, maternal characteristics, and survey-year fixed effects. Standard errors clustered at the child level are shown in parentheses. The F statistic is the Kleibergen-Paap statistic.

* and ** indicate statistically significant at the 10% and 5% levels.

C Additional Figures

C.1 Age Gradient: Robustness to Additional Controls

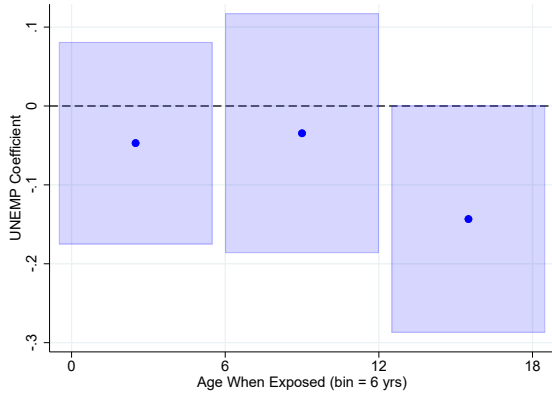
In the main text, Figure 2 documents the age gradient in the association between maternal unemployment and children’s adult wages using the baseline specification. Here, we assess whether this pattern is robust to (i) conditioning on family permanent income, (ii) adding proxies for maternal quality (educational attainment and AFQT scores), and (iii) correcting for measurement error in permanent income using our IV strategy.

Figure C1 presents the results. Panels (a) and (c) show OLS estimates that include age-specific permanent income controls (0-5, 6-11, 12-17) and maternal quality proxies. Panels (b) and (d) show IV estimates where permanent income is instrumented to correct for measurement error. In the IV specifications, we use a single permanent income measure for ages 0-17 rather than age-specific measures due to weak first-stage identification with multiple instruments. Because the OLS and IV specifications condition on different permanent income measures, they should be interpreted as targeting related but distinct conditional associations.

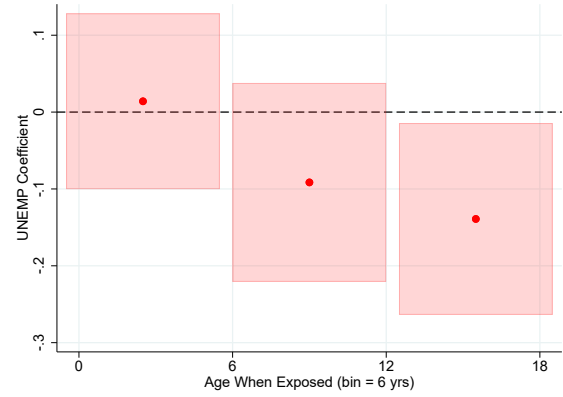
The age gradient is unchanged. In all specifications, maternal unemployment exposure during early childhood has little predictive power for adult wages, while exposure during adolescence remains a strong predictor of lower wages. The OLF pattern is also stable: early-childhood non-participation is associated with higher wages, while adolescent non-participation is associated with lower wages. These results confirm that neither permanent income nor maternal ability proxies, nor measurement error in permanent income, can account for the concentration of the association in adolescence.

Figure C2 presents the corresponding results for the age at labor force entry. Panel (a) reproduces the OLS estimates conditioning on age-specific permanent income, and panel (b) shows IV estimates. The association between adolescent exposure and earlier full-time labor-market entry is robust to both specifications.

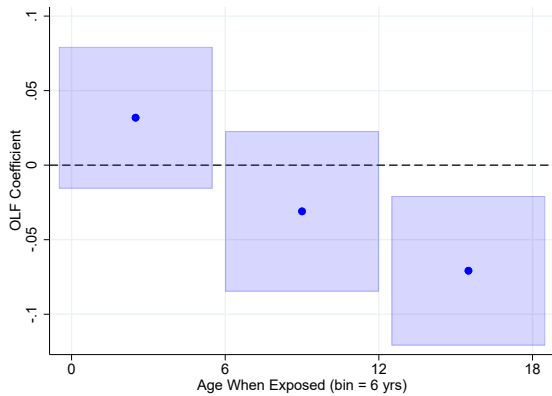
Figure C1: Association Between Maternal Employment Status and Wages by Childhood Stage: Additional Controls



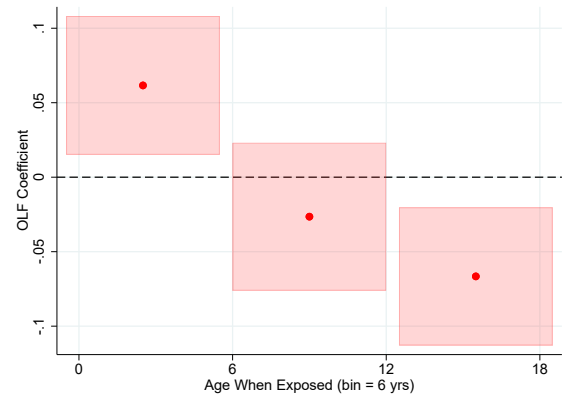
(a) Unemployment, OLS with PI and maternal quality controls



(b) Unemployment, IV



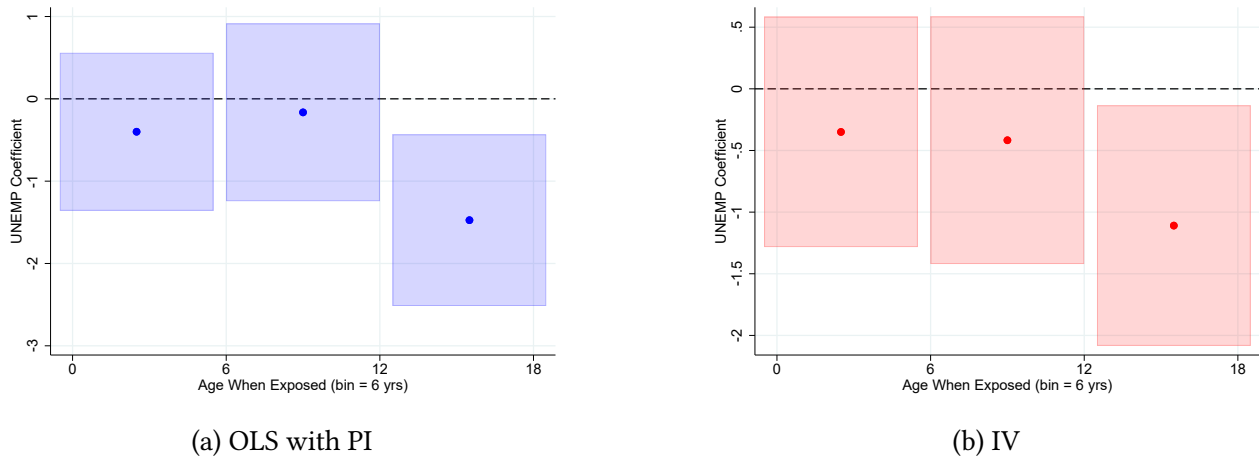
(c) Out of the labor force, OLS with PI and maternal quality controls



(d) Out of the labor force, IV

Note: This figure plots the estimated associations between maternal unemployment (panels a and b) and labor-force non-participation (panels c and d) and children's log wages, by age of exposure. Panels (a) and (c) show OLS estimates with age-specific permanent income controls (0-5, 6-11, 12-17) and maternal quality proxies (educational attainment and AFQT score quintiles). Panels (b) and (d) show IV estimates where permanent income is instrumented to correct for measurement error, using a single permanent income measure for ages 0-17 due to weak first-stage identification with multiple instruments. All specifications control for child demographics (race, sex, age polynomials, education) and maternal characteristics (marital status, primary earner status). The x-axis represents the child's age at exposure, grouped into 6-year bins. Point estimates are shown with 95% confidence intervals, with standard errors clustered at the individual level.

Figure C2: Association Between Maternal Unemployment and Age at Labor Force Entry: Conditioning on Permanent Income



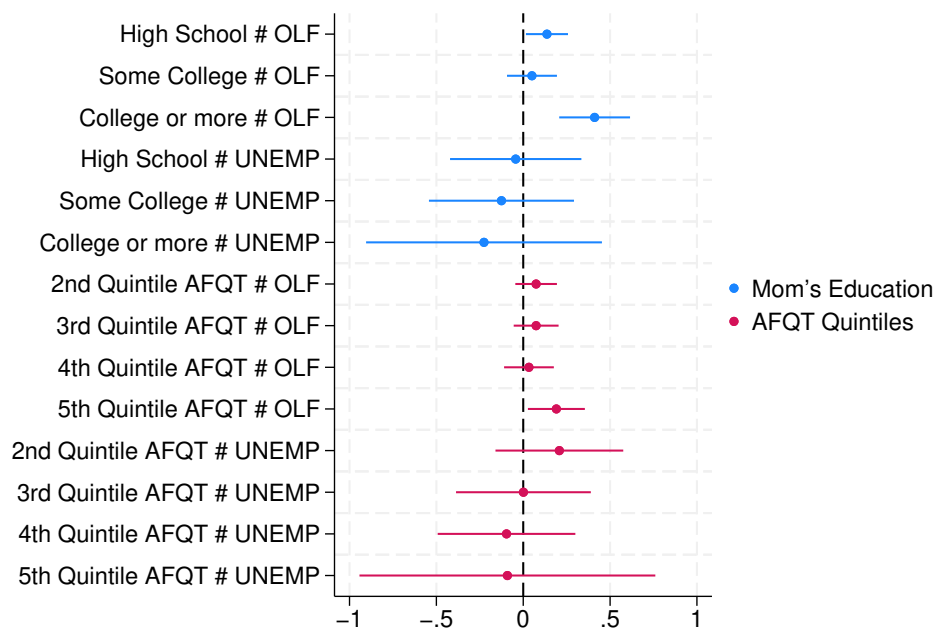
Note: This figure plots the estimated association between maternal unemployment exposure and children's age at labor market entry, by age of exposure. Panel (a) shows OLS estimates conditioning on age-specific permanent income (0-5, 6-11, 12-17). Panel (b) shows IV estimates where permanent income is instrumented to correct for measurement error, using a single permanent income measure for ages 0-17. Labor market entry is defined as the age at which the respondent first holds a full-time position (average weekly hours exceeding 35 hours), restricting attention to jobs starting at age 14 or later. Both specifications control for child demographics (race, sex, age polynomials), maternal characteristics (marital status, primary earner status), and maternal quality proxies (AFQT, maternal education). The x-axis represents the child's age at exposure, grouped into 6-year bins. Point estimates are shown with 95% confidence intervals, with standard errors clustered at the individual level.

C.2 Heterogeneous Effect by Mother’s Ability

We test for heterogeneity by interacting unemployment and labor force non-participation exposure with maternal education dummies and AFQT scores. Figure C3 plots the interaction coefficients.

The interaction terms are statistically insignificant, indicating no strong evidence of heterogeneity in the effects of maternal unemployment by maternal education or ability. However, point estimates suggest unemployment may be slightly more harmful for children of college-educated mothers, while labor force non-participation shows the opposite pattern—children of college-educated mothers who are out of the labor force have better outcomes than children of less-educated mothers who are out of the labor force. This asymmetry is consistent with differential selection: non-participation likely reflects deliberate childcare investment for college-educated mothers but discouraged-worker effects for less-educated mothers.

Figure C3: Heterogeneous Effect: Measures of Mother’s Ability



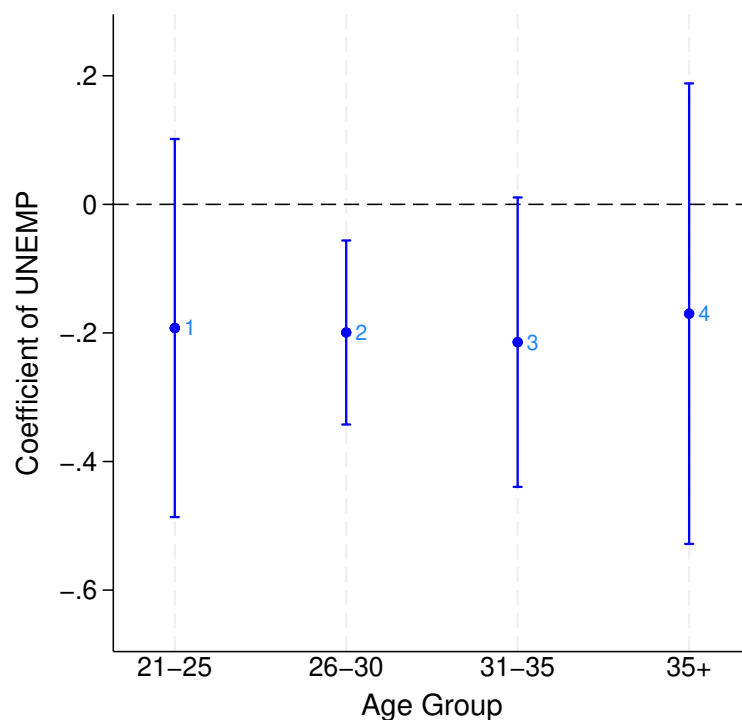
Note: This figure plots the coefficients from estimating equation 1 with children’s wages as outcomes, augmented with the interaction terms between our measures of exposure to maternal non-employment (OLF and UNEMP) and measures of mother’s cognitive abilities (education and AFQT quintiles). Each dot represents a point estimate of the effect of a variable displayed on the left-hand side on children’s future wages, with the confidence interval plotted around. The magnitudes of the point estimates and the confidence intervals can be determined by looking at the X-axis.

C.3 Persistent Effect of Unemployment Over Age Groups

We also check how the impact of maternal unemployment varies across different age groups. The goal is to determine whether the effects of experiencing unemployment during childhood persist as individuals grow older or if they diminish over time. We estimate the effect of unemployment by dividing the sample into four age groups: (i) 21-25 years, (ii) 26-30 years, (iii) 31-35 years, and (iv) 35+ years. We estimate using our specification corrected for measurement error and with proxies for the mother's quality.

Figure C4 presents the estimated coefficients across the four age groups, with confidence intervals represented by vertical error bars. The estimated effect of unemployment is negative across all age groups and has a similar magnitude, with no clear upward or downward trend over time. The coefficient is insignificant in some specifications because of the reduced number of observations.

Figure C4: Persistent Effect of Unemployment Over Age Groups



Note: This figure plots the coefficients of exposure to maternal unemployment on children's wages. Each dot represents a point estimate of the effect of unemployment at a given age group on children's future wages, with confidence intervals displayed around the estimates. The magnitude of the point estimates and the width of the confidence intervals can be interpreted by referring to the Y-axis.

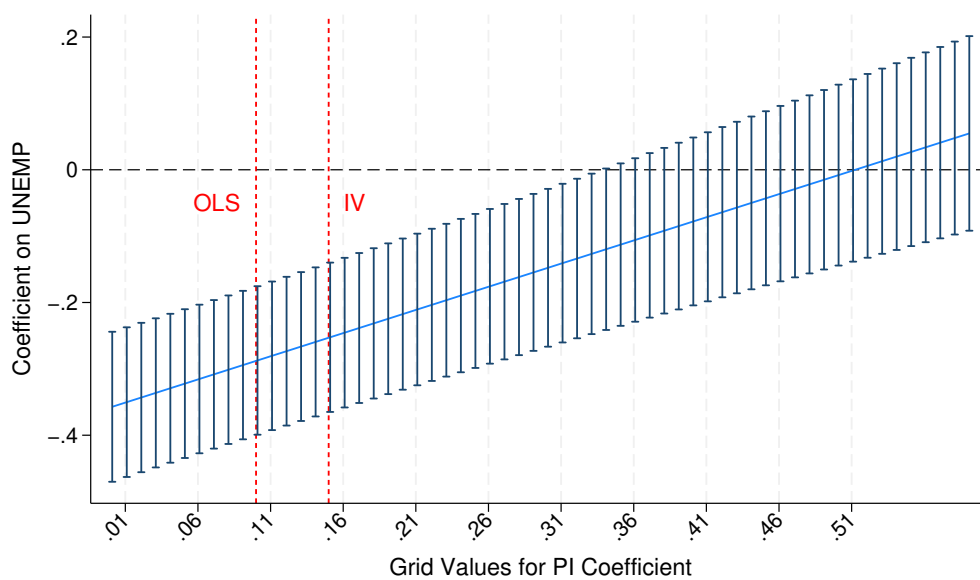
C.4 Assessing the Downstream Impact of Unemployment Beyond Income

Maternal unemployment impacts the outcome directly and through income. So, we control for a measure of permanent income to estimate the impact of unemployment not explained by the income channel. However, for that, it is crucial to estimate the coefficient on permanent income correctly. We deal with this concern first using instrumental variable estimation to correct for measurement error in permanent income. Additionally, we conduct a constrained regression where the coefficient on permanent income is fixed at different values within a pre-specified grid. This approach allows us to assess how the estimated impact of maternal unemployment changes as we impose different assumptions about the true value of the income coefficient.

Figure C5 shows that, if the true PI coefficient were large enough, the estimated impact of unemployment could approach zero—indicating that income fully accounts for any observed effects of unemployment. In particular, the graph plots the estimated coefficient on UNEMP (y-axis) as a function of different fixed values of the PI coefficient (x-axis). Two vertical lines represent the OLS and IV estimates of the PI coefficient.

For the estimated effect of unemployment to be fully accounted for by income (i.e., for the unemployment coefficient to reach zero), the true PI coefficient would need to be around 0.5. This value is five times larger than the OLS estimate and more than three times larger than the IV estimate. This suggests that income alone cannot fully explain the effects of unemployment unless one assumes an implausibly large coefficient on permanent income. In other words, unemployment likely has an independent impact beyond its effect through income. This highlights the importance of considering non-income channels when studying the consequences of unemployment.

Figure C5: Assessing the Downstream Impact of Unemployment Beyond Income



Note: This figure presents results from a constrained regression exercise where the coefficient on permanent income (PI) is fixed at different values along a predefined grid. By imposing these constraints, we examine how the estimated coefficient on unemployment (UNEMP) changes. This approach helps assess the extent to which income alone accounts for the observed impact of unemployment.